PRIMARY FIVE TERM ONE

<table>
<tr><td>THEME</td><td colspan="7">LIVIING TOGETHER IN UGANDA</td></tr>
<tr><td>TOPIC</td><td colspan="7">LOCATION OF UGANDA ON THE MAP OF EAST AFRICA</td></tr>
<tr><td>SUBTOPIC</td><td>SESSION</td><td>COMPETENCE</td><td>FACTS/CONTENT</td><td>METHODS</td><td>RELEVANCE</td><td>LESSON ACTIVITY</td><td>SCENARIO</td></tr>
<tr><td>DISTRICTS THAT FORM UGANDA</td><td>01</td><td>✓ Identifies districts and regions that form Uganda
✓ Describes highland and island districts</td><td><u>A district</u> is an administrative area at the level of Local Council Five (LC V) Uganda is divided into districts mainly to ease administration.
Chairperson LC V is the political head of a district.

Districts in Uganda are grouped into regions, and each region is divided into sub-regions according to the customs of the people in that area.

<u>Regions that form Uganda</u>

Northern region
Western region
Eastern region
Central region

<u>Sub-regions found in each region of Uganda.</u>
<table><tr><td>Region of Uganda</td><td>Sub-region</td><td>Districts</td></tr><tr><td>Central region</td><td>Buganda sub-region</td><td>-Kampala -Mpigi -Masaka -Mityana -Luwero -Kalangala -Nakasongola -Mubende -Rakai -Wakiso -Sembabule etc.</td></tr><tr><td rowspan="2">Northern region</td><td>Acholi sub-region</td><td>-Gulu -Kitgum -Agago -Pader</td></tr><tr><td>Karamoja sub-region</td><td>-Kotido -Abim -Moroto -Kaabong -Nakapiripirit etc</td></tr></table></td><td>Guided discovery

Guided discussion

Think pair share

Brain storming</td><td>To enable the learner to understand the geography of his environment</td><td>Identifying diffenrt districts in Uganda

Identifying neighbouring countries using cardinal ponts</td><td>Finding my home

Naming my home area

Tourism

School map

Study trips</td></tr>
</table>

					Lango sub-region	-Lira -Apac -Oyam -Dokolo etc					
					West Nile sub-region	-Arua -Nebbi -Zombo -Yumbe -Adjumani -Koboko					
				Western region	Bunyoro sub-region	-Hoima -Buliisa -Kibaale -Masindi					
					Tooro sub-region	-Kabarole -Kamwenge -Kyegegwa					
					Rwenzori sub-region	-Kasese -Bundibugyo -Ntoroko					
					Kigezi sub-region	-Kabale -Kisoro -Kanungu -Rukungiri					
					Ankole sub-region	-Mbarara -Ibanda -Kiruhura					
				Eastern region	Busoga sub-region	-Jinja -Iganga -Mayuge -Luuka -Buyende					
					Teso sub-region	-Soroti -Serere -Kumi -Amuria etc					
					Bugishu sub-region	-Mbale -Sironko -Bududa -Bulambuli					
					Sebei sub-region	-Kapchorwa -Bukwo -Kween					
					Bukedi sub-region	-Pallisa -Kibuku -Budaka -Tororo -Busia					

			REGIONS THAT MAKE UP UGANDA. MAP OF UGANDA SHOWING ALL POLITICAL REGIONS IN UGANDA Reasons why Uganda is divided into districts. To ease administration. To extend social services nearer to people. To create more job opportunities to people. Disadvantages of creating more new districts. It leads to separation of communities/ tribes. It leads to high administrative costs. It makes equal distribution of resources difficult. HIGHLAND DISTRICTS These are districts which are located in areas of high altitude. Such districts usually have many hills and mountains. Altitude is the height above the sea level. Examples of highland districts. Mbale Bundibugyo Kapchorwa Kasese Kabale Moroto Kisoro Rukungiri Ecnomic activities that are commonly carried out in highland districts. Tourism Mining Crop growing Problems faced by people living in highland districts. Landslides Poor transport network Severe soil erosion Difficulty in agricultural mechanisation. Possible solutions to the problems facing people living in highland districts. By rearing donkeys to use them as means of transport.				

			By constructing winding roads. By practising afforestation and reafforestation to control landslides. By terracing, contour ploughing and strip cropping to reduce soil erosion. Note: *Heavy rainfall is the major cause of landslides in highland areas.* *Landslides can be controlled in highland areas by planting more trees.* *Ground transport is very poor in highland areas because it is very difficult and expensive to construct roads in highland areas.* *Agricultural mechanisation is so poor in highland areas because the steep slopes make the use of tractors difficult.* *Road transport can be improved in highland areas by constructing winding roads.* *People living in Kapchorwa district rear donkeys mainly to use them as means of transport.* ISLAND DISTRICTS. An island is an area of land which is completely surrounded by a water body. The island districts in Uganda are found in Lake Victoria. Examples of Islands districts. Kalangala Buvuma Economic activities that are commonly carried out in Island districts. Fishing. Crop cultivation (oil palm growing, maize, banana, potatoes) Lumbering Trading. Tourism due to presence of Ssese forest, Ngamba island which is famous for Chimpanzees Benefits enyoyed by Island districts. They get easy access to water. They have fertile soils. They receive plenty of rainfall. Problems facing people living in Island districts. Poor road transport. Attacks from aquatic animals eg.crocodiles. Lack of hydro electricity Floods Poor social service delivery.				

			➢ Note*: -There is no hydroelectricity in Kalangala and because it is very expensive to fix electric poles in water of lake Victoria to these districts.* <u>Possible solutions to the problems facing Island districts of Uganda.</u> By using other sources of energy eg. solar energy By employing more social service providers. By providing better means of water transport. By building better social service centres in island districts. By paying attractive salaries to people who provide social services in island districts				
MAPS AND PICTURES	02	✓ Gives the meaning of a map and a picture ✓ Illustrates maps and pictures of different objects ✓ Describe elements of a map Describes map symbols	MAPS AND PICTURES <u>A map</u> is a drawing/ representation of an object as seen from above. Maps show how objects look like when viewed directly from above. We use symbols to help us understand and interpret a map of a detailed area. The boundary of a map determines the size and shape of the map. <u>A picture</u> is a representation of an object as seen from above. All features on a picture can be clearly seen as real objects. Pictures are more detailed than maps. <u>Maps and Pictures of different objects</u> Pot Tree Plate House Basin Chair Cup Hut Car Table Bottle <u>Similarity between maps and pictures.</u> • Both are representations of objects. <u>Types of maps.</u> Political maps. These maps show boundaries of villages, counties, districts and countries.	Guided discovery Guided discussion Think pair share Brain storming		Drawing the map of Uganda showing major lines of latitude and longitude Drawing maps and pictures Measuring the distance of places	Drawing images Taking pictures on a party Capturing videos Measuring

Topographic maps. These maps show physical features/ landforms of a given area .eg. mountains etc.

Flow-line maps. These show movement of people, goods, animals etc.

Thematic maps. These show various social and economic themes eg. trade, energy.

<u>Importance of maps.</u>
They are used for locating places.
They help travelers to plan for routes to take while on their journeys.
They help people to know the relief, climate and vegetation of different areas.

<u>ELEMENTS OF MAP</u>

A compass direction
A scale
A title/heading
A frame
A key
<u>Importance of different elements of a good map.</u>

Element	Importance
A compass direction	It helps to show the direction of places on a map. A map reader may fail to know the direction of places shown on the map if he/she reads a map without a compass direction.
A title/ heading	It helps one to know what the map is all about. One who reads a map without a title may fail to know what the map is all about.
A key	It helps a map reader to interpret symbols used on a map. A key gives detailed information about the map.

	A map reader may fail to interpret/ know the meaning of symbols used on a map if he/she reads a map without a key.
A scale	It helps a map reader to calculate the actual ground distance on a map. A map reader may fail to calculate the actual ground distance between places on the map if he/she reads a map without a scale.
A frame	It shows the extent of the area represented by the map.

MAP SYMBOLS
These are features that are used to represent real objects on a map.

<u>Why symbols are used on maps.</u>
To avoid congestion on the map.
To make map reading easy.
To ensure neatness of a map.

<u>Common symbols used on maps.</u>

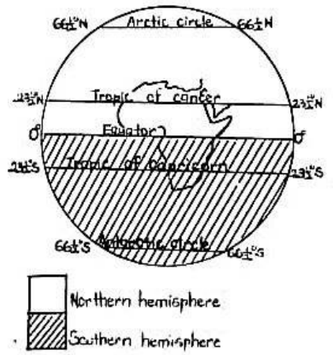
<u>A COMPASS DIRECTION</u>
<u>A compass</u> is an instrument used to find direction of places.
A drawn compass is called a compass rose.

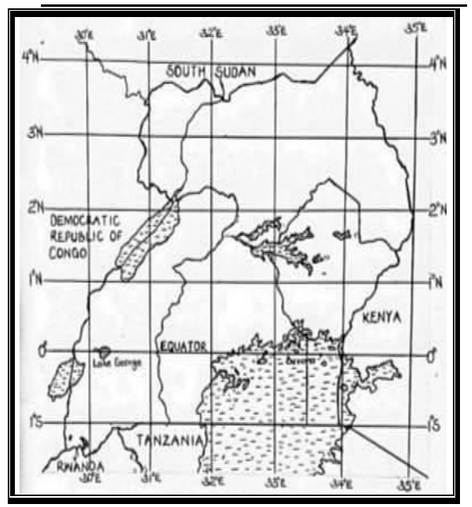
<u>People who use a compass.</u>
Rally drivers
Mountain climbers
Soldiers.
Tourists
Sailors
Pilots
Scouts and girl guides.

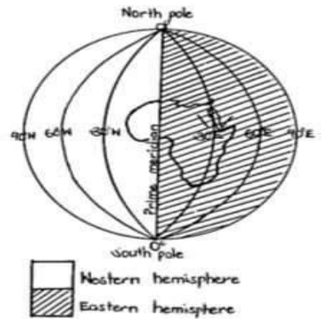
<u>A compass direction.</u>
<u>A compass direction</u> is a symbol used on a map to show the direction of places.

			The compass points are divided into the cardinal points, semi-cardinal points (secondary points) and tertiary points. <u>Cardinal points</u> are the four major points of a compass. These include; North South West East <u>Secondary points</u> are the directions that lie midway of cardinal points. Semi-cardinal points lie at 45˚from the cardinal points. These include; South East South West North East North West <u>A SCALE.</u> <u>A scale</u> helps a map reader to calculate the actual ground distance on a map. A map reader who reads a map without a scale may fail to calculate the actual ground distance on a map. A scale is only found on accurate maps. <u>A sketch map</u> is a map which is not drawn to scale while <u>an accurate map</u> is a map which is drawn to scale. <u>Types of scales.</u> Linear scale. This is the type of scale drawn using lines that are divided into equal parts. *Example;* *Find the actual ground distance between town Q and town S if there is a distance of 6cm between the two towns on the map.* *<u>Solution;</u>* *1cm = 100km* 6cm = (6x100) km				

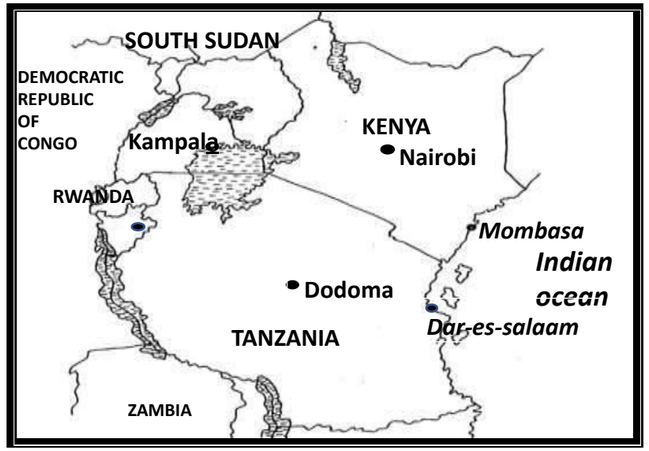
			= 600km Therefore the actual ground distance between Town Q and Town S is 600km. (ii) Representative scale/ fractional scale. This is the type of scale written as a mathematical fraction. It can also be written as a ratio.eg. $\frac{1}{100km}$ or 1:100km. (iii) Statement scale. This type of scale can be given as a statement or words. Eg.1cm on a map represents 100km on the ground				
LOCATING UGANDAUSING GRIDREFERENCE SYSTEM	03	✓ Ways of locating places physically. ✓ Explains ways of locating places on a map Describes lines of latitude	LOCATING UGANDAUSING GRIDREFERENCE SYSTEM Ways of locating places By using a compass. By using neighbouring places. By using landmarks. Ways of locating places on a map Using neighbouring places. Using physical features. Using lines of latitude and lines of longitude (grid reference system) <u>GRID REFERENCE SYSTEM</u> This is the method of locating places using lines of latitude and lines of longitude. <u>Grid</u> refers to the network of horizontal and vertical lines used as references coordinates to locate places on a map. <u>LINES OF LATITUDE.</u> These are imaginary lines drawn on maps or globe from East to West. The general name for all lines of latitude is <u>parallels</u> because they do not meet at any point. <u>Latitude</u> is the distance in degrees North or South of the equator. Lines of latitude help in determining climate of places. Lines of latitude help us to tell how far a place is North or South of the Equator. <u>Major lines of latitude.</u>	Guided discovery Guided discussion Think pair share Brain storming		Identifying latitudes and longitudes	Finding my home Naming my home area Tourism School map Study trips

			• The Equator (0°) • Arctic circle ($66½^oN$) • Tropic of Cancer ($23½^oN$) • Antarctic circle ($66½^oS$) • Tropic of Capricorn ($23½^oS$)  <u>THE EQUATOR</u> *The Equator divides the world into two equal parts (hemispheres)* *The Equator is marked 0° because it is the starting point for all latitude readings.* *The Equator is the most important line of latitude because it crosses the world at the centre.* *All districts which are crossed by the equator lie in both the Northern and Southern hemisphere.* *Note:* *A globe is the model of the earth* *The Equator crosses lake George and lake Victoria in Uganda.* *<u>A hemisphere</u> is a half part of the world as divided by the equator or prime meridian.* *The area between the Tropic of cancer and the Tropic of Capricorn is called the <u>Tropical</u> <u>region</u>.* Districts crossed by the equator Ibanda Kasese Mukono Mpigi Wakiso Namayingo Mayuge Kalungu				

			Bukomansimbi <u>Importance of lines of latitude</u> They help to determine the climate of a place. Lines of latitude and lines of longitude are used to locate places on a map. <u>Location of the Equator on the map of Uganda</u> *<u>Practical activity;</u>* *Use the political map of Uganda and identify all districts that;* *(i) Are crossed by the Equator.* *(ii) Completely lie in the Northern hemisphere* *(iii) Completely lie in the Southern hemisphere.* *(iv) Lie in both the Northern and the Southern hemisphere*  				

			UGANDA'S NEIGHBOURS Uganda is bordered by five (5) countries. These include; Kenya in the East Democratic Republic of Congo in the Tanzania in the South West South Sudan in the North Rwanda in the South West.				
UGANDA AS A LANDLOCKED COUNTRY	04	✓ Describes line of longitude ✓ Explains the use of Prime meridian to tell international time ✓ Locates Uganda using the grid reference system ✓ Discusses rotation and revolution of the earth (causes and effects) ✓ Describes Uganda neighbours	) <u>LINES OF LONGITUDE.</u> <u>Lines of longitude</u> are imaginary lines drawn on maps and globe from the North pole to the South pole They are also known as <u>meridians.</u> Lines of longitude meet at the poles of the earth. <u>Major lines of longitude.</u> The Prime meridian/ Greenwich meridian (0°) The International dateline. (180^oE or W)  *<u>The Prime Meridian/ Greenwich Meridian</u>* *The Prime meridian is also called the <u>Greenwich</u> <u>meridian</u> because it crosses Greenwich town in London.* *The Prime meridian is marked 0° because it is the starting point for all longitude readings. The Prime meridian divides the world into the Eastern and Western hemisphere.* *All countries crossed by the Prime meridian lie in both the Eastern and Western hemisphere.* *<u>Note;</u>* • *The Prime meridian helps in telling international time.* • *The International dateline separates one day from the next day.* *e.g. East of the International dateline may be Wednesday while West of it is Tuesday.*	Guided discovery Guided discussion Think pair share Brain storming		Locating places using lines of longitude Identifying neighbouring countries of Uganda Drawing the map of Uganda showing her neighbours	Finding my home Naming my home area Tourism School map Study trips

			<u>Using the Prime meridian to tell the international time.</u> A time zone is an area/region with the same standard time. Different regions of the world have different time zones according to the distance from the Prime meridian. Places which are in the same time zone have the same standard time. e.g. Uganda, Kenya and Tanzania are in the same time zone, and so have the same standard time. (East Africa standard time) East Africa lies at longitude 45° while Rwanda and Burundi are 30°E of the Greenwich meridian. <u>Note:</u> *The rotation of the earth on its axis causes days and nights.* *The earth makes one complete rotation of 360° in one day (24hours).* *Therefore, it rotates through an angle of 15° in every 1hour (60min).* *Every 15° E or W of the Greenwich meridian is a time zone.* *When you travel 15° westwards, you lose an hour while travelling 15° eastwards makes you gain an hour.* *We usually add (+) hours for places which are in the East and subtract (-) the time when finding the time for places which are in the West.* *Changing of the time to and from 24hr clock (by either adding or subtracting 12 hrs) affects the units in which the time is given.(ie. From am to pm and viceversa).* *Time in each time zone is calculated basing on the Greenwich mean time (GMT) which is at 0° longitude.* <u>*Example 1:*</u> **1**. Find the time in East Africa which is 45° if it is 1:00pm at GMT. <u>*Solution,*</u> 15°=1hr <u>OR:</u> 45$_{o}$=(^{45}) hrs. =3hrs Time in East Africa=1:00pm <u>+3.00hrs</u> <u>4:00pm</u> <u>*Therefore, time in East Africa is 4:00pm*</u> <u>*Therefore, time in East Africa is 4:00pm.*</u> <u>Locating Uganda using the grid reference system.</u> Uganda is located between latitude 4°N and 1°S and longitude 29°E and 35°E of the Greenwich meridian. The farthest point in the North is Zulia while Kisoro is the farthest in the South.			

			Ishaka is the farthest town in the West while Amudat is the farthest in the East. Location of Uganda using the grid reference system <u>Location of Uganda on the map East Africa</u>  <u>Countries that neighbour Uganda.</u> Country / Capital City / Current President Kenya / Nairobi / H.E Uhuru Kenyatta Tanzania / Dodoma / H.E John Pombe Magufuli DRC / Kinshasa / H.E Felix Tshisekedi South Sudan / Juba / H.E Salva Kiir Rwanda / Kigali / H.E Paul Kagame				
	05	✓ Gives the meaning of a landlocked country	UGANDA AS A LAND LOCKED COUNTRY <u>A land locked country</u> is a country without a coastline. Uganda is a land locked country because she has no coastline.	Guided discovery		Identifying landlocked countries in East Africa	Tours

		✓ Locates landlocked countries that boarder Uganda ✓ Locates sea ports that handle Uganda's imports and exports ✓ Explains problems associated with landlocked countries ✓ Describes the steps landlocked countries can take/have to solve the some of the above challenges ✓ Gives the meaning of nonland locked countries ✓ Locates nonland locked countries that neighbor Uganda ✓ Explains the benefits enjoyed by nonland locked countries Explains reasons why Uganda should cooperate with her neighbors	Uganda usually uses her neighbours' sea ports when importing and exporting her goods. <u>Land locked countries that border Uganda.</u> Rwanda South sudan <u>Seaports that handle Uganda's imports and exports.</u> Port Mombasa in Kenya Port Dar-es-salaam in Tanzania Port Matadi in Democratic Republic of Congo Note: *A Seaport is a a place at the coast where ships anchor.* *Uganda mostly imports her goods through port Mombasa because port Mombasa is nearer to Uganda than other sea ports.* <u>Problems faced by land locked countries in Uganda.</u> High taxation on overseas goods/imports. Delay of overseas goods in transit. High transport costs when importing goods. There is limited trade with the rest of the world. High prices for imported goods. Note: *Smuggling is the illegal importation and exportation of goods.* *Smuggling reduces market for locally manufactured goods.* *It may also lead to importation of low quality goods into the country.* *Imports are goods that are brought in a country from other countries eg. Vehicles Uganda imports from Germany.* *Exports are goods a country sells to other countries.* *eg.food Uganda sells to South Sudan.* <u>Steps land locked countries can take/ have taken to solve some of the above challenges.</u> <u>(How Uganda can solve the problems she faces due to her location)</u> By using air transport when importing goods. By using various sea ports when importing goods. By promoting domestic industrial production. By joining common markets in the region.eg.East African Community (EAC) By improving security along the high ways.	Guided discussion Think pair share Brain storming		Identifying Uganda's neighbours	Theft Sneaking things into the school

			NON-LAND LOCKED COUNTRIES. A Non-land locked country is a country which has a coastline. Non-landlocked countries have their own sea ports that handle their imports and exports. Non-landlocked countries that neighbour Uganda. Kenya Democratic Republic of Congo Tanzania Major sea ports of Uganda's neighbours. Non land locked countries | Sea Ports Kenya | Port Mombasa Tanzania | Port Dar es salaam Democratic Republic of Congo | Port Matadi Benefits enjoyed by Non-land locked countries. They pay less taxes on their imported goods. They earn income through taxing goods for land locked countries that pass through them. They can easily trade with other countries. Note: ✓ *Hinterlands are areas in a country that are far away from the coast or main cities.* Reasons why Uganda should cooperate with her neighbours To widen market for her goods. To use her neighbours' sea ports to handle her exports and imports. To get goods which she can't produce locally?				
TOPIC	PHYSICAL FEATURES OF UGANDA						
SUBTOPIC	SESSION	COMPETENCE	FACTS/CONTENT	METHODS	RELEVANCE	LESSON ACTIVITY	SCENARIO
INTRODUCTION TO PHYSICAL FEATURES	06	✓ Gives that meaning of physical features ✓ Describes examples of physical feature	**INTRODUCTION TO PHYSICAL FEATURES** Physical features are land forms of the earth's surface that give it shape. Examples of physical features.	Guided discovery	It enables the learner to understand his or her environment	Identifying physical features in Uganda	Tourism

		✓ Explains the types of physical feature ✓ Describes Block mountains in Uganda	Mountains. Hills. Lakes. Rivers. Plateau. <u>Types of physical features.</u> Drainage features. Relief features <u>Relief features.</u> ***<u>Relief features</u>*** are landforms identified by their altitude. <u>Relief</u> is the physical appearance of the land scape. <u>Altitude</u> is the height above the sea level. **<u>Examples of relief physical features.</u>** Mountain. Rift valley. Plateau. Valleys. Hills. Plains. ***<u>Drainage features.</u>*** These are landforms that contain water. **<u>Examples of drainage features.</u>** Lakes. Seas. Rivers. Oceans. <u>A mountain</u> is a high large piece of land usually higher than a hill. **<u>Examples of mountains in Uganda.</u>**	Plains. Rift valleys. Seas. Oceans. Valleys.	Guided discussion Think pair share Brain storming		Differentiating drainage and relief features Drawing the structure of a block mountain	Nature of the land

Mount Rwenzori.	Mount Mufumbiro
Mount Zulia.	Mount Napak.
Mount Elgon	Mount Kadam.
Mount Morungole	Mount Moroto.

Types of mountains in Uganda.

Volcanic
mountains.
Block mountains

i) BLOCK MOUNTAINS

These mountains are also called horst mountains.
Block mountains were formed as a result of faulting.

Faulting is the cracking/ breaking of rocks in the earth's crust.

The central block between the two faults is lifted up by compressional forces forming a block mountain.

Forces that led to the formation of a block mountain
Compressional
forces
Tensional forces.

Example of a block mountain in Uganda

Mountain Rwenzori

A diagram showing the formation of a block mountain due to Compressional forces.

MOUNT RWENZORI
It is the highest mountain in Uganda.

			Mountain Rwenzori is located in Western Uganda. It is surrounded by districts like Kasese, Kabalore and Bundibugyo. It forms a boundary between Uganda and Democratic Republic of Congo. It has many peaks ie Margherita as the highest and Stanley as the second highest. It is called a range because it has many peaks. Margherita is snowcapped throughout the year because it is above the snow line. Henry Morton Stanley was the first European explorer to see mountain Rwenzori. He named it the "mountains of the moon" because of the snow on its peak which glitters like a moon. Mountain Rwenzori is a source of rivers like River Mubuku, river Sebwe, River Nyamwamba, River Mpanga. These rivers flow from molten snow on mountain rivers. Tribes that live on the slope of mountain Rwenzori. Bakonzo. Bamba Crops commonly grown on the slopes of mountains. Arabica coffee Bananas.				
MOUNTAINS IN UGANDA	07	✓ Describes volcanic mountains in Uganda ✓ Describe volcanicity Locates major mountains on the map of Uganda	VOLCANIC MOUNTAINS. These are mountains formed as a result of volcanicity. Volcanicity is the process by which magma is forced onto the earth's surface through a vent. Volcanicity can also be called volcanic activity or volcanic eruption. A volcano is a mountain with a vent through which magma is forced through the earth crust and onto the earth's surface. Magma refers to the molten rock in the earth crust.	Guided discovery Guided discussion		Identifying volcanic mountains Drawing the formation of	Water scarcity Project making

			Lava refers to the molten rock on the earth's surface. **A diagram showing the formation of a volcanic mountain** **<u>Types of volcanoes</u>** 	Type of volcano	Description	Example(s)
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Active volcanoes	These are mountains that have erupted in the recent past and still show signs of erupting.	Mount Mufumbiro
Dormant volcanoes.	These are mountains that have not erupted in the recent past but still show signs of erupting.	Mount Moroto.
Extinct volcanoes	These are mountains that do not show any signs of erupting.	Mount Elgon Mount Napak Mount Morungole Mount Zulia Mount Kadam	 **<u>Importance of volcanicity</u>** It leads to formation of fertile volcanic soils which are favourable for crop growing. eg.Arabica coffee is mainly grown on the slopes of mountain Elgon due to the presence of fertile soils. Volcanic mountains attract tourists who bring in income. **<u>Dangers of volcanicity.</u>** It leads to death of people and animals. It leads to destruction of property.	Think pair share Brain storming		volcanic mountains Identifying types of volcanic mountains	Soil erosion Wildlife migration

			It leads to air pollution. **<u>Other features formed as a result of volcanicity.</u>** Crater lakes. Calderas Inselbergs *<u>An inselberg</u> is an isolated hill that stands above the general level of the surrounding land eg.Labwor hill in Karamoja, Musaijamukuru hill in Hoima, Osukuru hill in Tororo.* *These inselbergs attract tourists who bring in income and are also a major source of minerals eg. The Osukuru hills which provide limestone used for making cement.* Lava dammed lakes Hot springs /geysers eg. Kitagata hotsprings in Bushenyi, Sempaya hotsprings in Bundibugyo. *Hot springs attract tourists and also help in generation of geothermal energy.* <u>MOUNTAIN ELGON.</u> It is located in Eastern Uganda. It is the second highest mountain in Uganda. It is locally known as mountain <u>Masaba.</u> Its highest peak is <u>Wagagai.</u> It forms a boundary between Uganda and Kenya. The Bagishu and Sabiny live on slopes of mount Elgon. <u>Joseph Thomson</u> was the first European explorer to see mount Elgon. Mount Elgon national park is located on mount Elgon. <u>Crops commonly grown on the slopes of mount Elgon</u>. Arabica coffee Bananas Wheat Maize <u>Factors that favour the growth of arabica coffee on the slopes of mount Elgon</u> Presence of volcanic fertile soils. Presence of heavy rainfall. Presence of cool temperature. Note: *Kapchorwa district is the leading producer of wheat in Uganda.*				

			People living in Kapchorwa rear donkeys mainly to use them as means of transport since their district experiences poor transport network. <u>Districts that share part of mount Elgon</u> Mbale Sironko Kapchorwa Manafwa <u>Rivers that originate from mount Elgon.</u> River Manafwa River Malaba River Mpologoma River Nzoia in Kenya <u>MOUNT MUFUMBIRO.</u> It is located in the South Western part f Uganda. Its highest peak is <u>Muhavura.</u> Other peaks include mgahinga and Sabinyo. It is shared by Uganda, Rwanda and Democratic Republic of Congo. It forms a range of mountains in the Kigezi sub-region in South Western part of Uganda. The Mufumbiro range is a home for the mountain gorillas in Bwindi and Mgahinga national park. Mount Mufumbiro mainly occupies Kabale, Kisoro and Kanungu districts. The area around mount Mufumbiro was referred to as "<u>the Switzerland of</u> <u>Africa</u>" by <u>Winston Churchill</u> because it has features which are similar to those of the Alps mountains in Switzerland. <u>Tribes that live on the slopes of mount Mufumbiro.</u> Bakiga Bafumbira. <u>Crops mainly grown on slopes of mount Mufumbiro</u> Potatoes Sorghum Vegetables Pyrethrum ➢ Note: *The high population/ over population in Kigezi sub-region has led to the division of land into small plots in this area (Land fragmentation)*				

			Land fragmentation hinders agricultural mechanisation and also leads to low food production. <u>MOUNT MOROTO.</u> It is located in the North Eastern part of Uganda. Its highest peak is called <u>Sokdek.</u> Sogolomon is the second highest peak of mount Moroto. The Karimojong tribe mainly lives on the slopes of mount Moroto. The Karimojong mainly carry out pastoralism. The area around mount Moroto receives very little rainfall because it receives dry winds from the North East.				
	08	✓ Explains the importance of hill and mountains ✓ Illustrates the formation of relief rainfall (orographic rainfall) ✓ Describes the economic activities commonly carried out in mountainous areas ✓ Explains the disadvantages of mountains ✓ Describes the problems faced by people living of slopes of mountains ✓ Explains why transport is very poor in highland areas Explains why agricultural mechanization is difficult/ poor mountainous area	**<u>Importance of hills and mountains</u>** They help in rain formation ie.Relief rainfall. They have fertile soils which support crop cultivation. They attract tourists who bring in income. They have cool temperatures which favour dairy farming. They are used by telecommunication companies to place on masts. Some mountains are a source of rivers. Some mountains are a source of minerals eg.Osukuru hills in Tororo which provide limestone. They are a source of building materials eg.stones Some mountains form natural boundaries between countries. Eg-Mountain Rwenzori creates a natural boundary between Uganda and DRC. Mountain Elgon creates a natural boundary between Uganda and Kenya. Mountain Mufumbiro creates a natural boundary between Uganda and Rwanda. A diagram showing the formation of Relief (Orographic rainfall) **<u>Economic activities commonly carried out in mountainous areas.</u>** Crop growing Lumbering Stone quarrying Tourism Mining **<u>Disadvantages of mountains.</u>** They hinder agricultural mechanisation.	Guided discovery Guided discussion Think pair share Brain storming		Identifying importance of hills mountains Drawing the formation of relief rainfall Identifying advantages and disadvantages of mountains	Water scarcity Project making Soil erosion Wildlife migration

			They make road and railway construction difficult. Some volcanic mountains erupt and destroy property. They are greatly affected by landslides and soil erosion. **<u>Problems faced by people living on slopes of mountains.</u>** Severe soil erosion. Much coldness. Landslides due to heavy rainfall. Poor agricultural mechanisation. Poor road transport. **<u>Why transport is very poor in highland areas.</u>** It is difficult and expensive to construct roads in mountainous areas. Note: *People living in highland areas (Kapchorwa) rear donkeys to use them as means of transport.* *Heavy rainfall received in highland areas is the major cause of landslides.* *Landslides can be reduced in highland areas through planting trees.* *Road transport can also be improved in highland areas by constructing winding roads.* *The government has relocated people of Bududa to safer plains and plateaus to save them from landslides.* ***<u>Why agricultural mechanisation is poor in mountainous areas.</u>*** *The steep slopes make the use of tractors difficult and very expensive.*				
THE RIFT VALLEY	09	✓ Gives the meaning of a rift valley ✓ Describes and locates the arms of the rift valley and their extent on the map of East Africa ✓ Illustrates the formation of the rift valley ✓ Describes the features formed in a rift valley	**<u>THE RIFT VALLEY</u>** <u>A rift valley</u> is a long and wide depression on the earth's surface with steep sides. The steep sides of a rift valley are called <u>escarpments.</u> The rift valley was formed as a result of <u>faulting.</u> The width of the Great rift valley varies from 30 to 100km. <u>Arms of the rift valley.</u> Western rift valley (runs through Uganda) The Eastern rift valley (runs through Kenya and Tanzania) <u>Note:</u> -*Albertine escarpment is the major escarpment of the Western rift valley.* *Crude oil was discovered in the Albertine region.* **A diagram showing the formation of a rift valley.**	Guided discovery Guided discussion Think pair share		Identifying rift valley arms in Uganda Drawing the formation of a rift valley	Water scarcity Project making Pit digging

		✓ Explains the economic activities done in a rift valley ✓ Describes the importance of the rift valley ✓ Explains the dangers/ disadvantages of the rift valley	**<u>Features found in the rift valley.</u>** Lakes eg,lake Albert, lake Edward and lake George. Rivers Valleys Hills **<u>Economic activities done in Rift valley areas</u>** Crop cultivation Fishing Mining Animal rearing Tourism **<u>Importance of the rift valley.</u>** It promotes tourism. It has lakes which promote fishing. It promotes wild life conservation. It has plenty of pasture for animal rearing. It forms natural boundaries between countries eg. The Western rift valley forms a natural boundary between Uganda and Democratic Republic of Congo. Lakes found in the rift valley are a source of minerals eg.lake Katwe which provides salt. **<u>Dangers / disadvantages of the Rift valley.</u>** It is affected by soil erosion. It hinders road and railway construction. It is greatly affected by floods. It experiences very high temperatures due to low altitude.	Brain storming		Identifying features in the rift valley	Soil erosion Wildlife migration
THE PLATEAUS/PLATEAUX AND PLAINS	10	✓ Gives the meaning of a plateau ✓ Describes features found on a plateau ✓ Describes the economic activities carried	**<u>PLATEAUS / PLATEAUX PLAINS</u>** <u>A Plateau</u> is a raised flat topped piece of land. The plateau covers the largest part of Uganda. It lies between 200m and 2000m above the sea level. **<u>Features found on a plateau</u>** Lakes Streams	Guided discovery Guided discussion		Describing plateaus	Farming

		out in plateau areas ✓ Explains the importance of plateaus ✓ Illustrate the formation of frontal/ cyclonic rain in plain areas ✓ Locates the water bodies on the central plateau of Uganda	Hills Rivers Valleys **<u>Economic activities carried out in plateau areas.</u>** Crop cultivation Mining Industrialisation. Tourism Animal rearing Fishing Lumbering **<u>Importance of plateaus.</u>** They have fertile soils for crop growing. They promote wildlife conservation. They have plenty of pasture for animal rearing. They promote tourism. The help in formation of cyclonic rainfall. **<u>A diagram showing the formation of Cyclonic / Frontal rainfall</u>**	Think pair share Brain storming		Identifying features on the plateau	Charcoal burning Games and sports
LAKES IN UGANDA	11	✓ Gives the meaning of a lake ✓ Describes major lakes in Uganda ✓ Locates the major lakes and rivers on the map of Uganda ✓ Describes the types of lakes Describes the examples of permanent lakes in Uganda	**<u>LAKES IN UGANDA</u>** <u>A lake</u> is a mass of non-flowing water on the earth's surface. <u>Or</u>: A lake is a depression / hollow on the earth's surface in which water collects. <u>Or</u>: A lake is a large area of water that is surrounded by land. <u>Examples of major lakes in Uganda.</u> Lake Victoria. Lake Kyoga. Lake Nakivali Lake Albert. Lake Edward. Lake Bisina Lake Mburo. Lake George. Lake Kijanebalola Lake Bunyonyi. Lake Wamala. Lake Opeta. Lake Katwe. Lake Kwania <u>Types of lakes.</u> Permanent lakes. Seasonal lakes.	Guided discovery Guided discussion Think pair share Brain storming		Describing lakes Identifying examples of lakes in Uganda	Study tours vacations Swimming

<table>
<tr>
<td></td>
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<td></td>
<td><u>Permanent lakes</u> are lakes which contain water throughout the year.

<u>Examples of permanent lakes in Uganda.</u>
Lake Victoria.
Lake Edward.
Lake Mburo.
Lake George.
Lake Albert.
Lake Kyoga

<u>Seasonal lakes</u> are lakes which dry up in the dry season and get water in the wet season.</td>
<td></td>
<td></td>
<td>Fishing project

Games and sports</td>
</tr>
<tr>
<td></td>
<td>12</td>
<td>✓ Describes basin lakes
(Lake Victoria and lake Kyoga)</td>
<td><u>BASIN LAKES.</u>

<table>
<tr><th>Description</th><th>Examples</th></tr>
<tr><td>These are also called basin lakes/ down warped lakes.
Basin lakes were formed as a result of <u>down warping.</u>
<u>Characteristics of basin lakes</u> They have irregular shapes.
They are wide.
They are usually shallow.
They have inlet and outlet rivers.
They have fresh water.</td><td>• Lake Victoria
• Lake Kyoga
• Lake Wamala
• Lake Mburo
• Lake Kwania
• Lake Bisina
• Lake Opeta
Note: -Basin lakes have fresh water because they have outlet rivers.</td></tr>
</table>

<u>LAKE VICTORIA.</u></td>
<td>Guided discovery

Guided discussion

Think pair share

Brain storming</td>
<td></td>
<td>Describing lakes

Identifying types of lake formations in Uganda</td>
<td>Study tours

vacations

Swimming

Fishing project

Games and sports</td>
</tr>
</table>

			Its local name is Nalubaale (Home of gods) in Uganda. It's locally known as Sango in Kenya and Nyanza in Tanzania. It was formed as result of down warping. It is the largest fresh water lake on the Central plateau of Uganda. It is referred to as an inter-territorial lake because it is shared by three countries in East Africa. It has inland ports which promote inter-territorial trade through handling the goods of the three East African countries. John Hanning Speke was the first European explorer to see Lake Victoria. John Hanning Speke named it Lake Victoria after Queen Victoria of England by then. Kalangala and Buvuma districts are completely surrounded by Lake Victoria. ***Inland ports on Lake Victoria.*** **Inland ports on Lake Victoria in Uganda.** Port Bell. Port Bukakata Port Jinja NB. *A port is a place on a water body where water vessels load and off loads.* **LAKE KYOGA.** It is a fresh water lake. It was formed by down warping. It is located on the Central plateau of Uganda. It is the swampiest lake in Uganda. This is because it is very shallow. Lake Kyoga is shallow because it is highly silted. The Victoria Nile flows from Lake Victoria towards lake Kyoga. This natural evidence proves that Lake Victoria is on a higher altitude than lake Kyoga. Inland ports of Lake Kyoga. Port Lwampanga. Port Namasala. Inlets of lake Kyoga. Victoria Nile. River Sezibwa. River Kafu.				

	13	✓ Gives the meaning of rift valley lakes ✓ Describes rift valley lakes (lake albert, George and lake albert)	RIFT VALLEY LAKES **Description.** Rift valley lakes are lakes which lie on the floor of the rift valley and were formed as a result of faulting. Characteristics of rift valley lakes ❖ They are long and narrow (oblong) ❖ They are deep. ✞ They are salty. ✞ They have steep sided shores. ✞ Most of them have no outlet rivers. Why rift valley lakes are salty. ✞ They have salty basement rocks. (because they experience a lot of evaporation) ✞ They have no outlet rivers. Note:-*Lake Katwe lies on the floor of the riftvalley but it is not considered to be a riftvalley lake because it was formed as a result of volcanicity.* **Examples** ✞ Lake Albert ✞ Lake George ✞ Lake Edward LAKE ALBERT. It is the second largest lake in Uganda. It was formed by faulting. It forms a natural boundary between Uganda and Democratic Republic Congo. Its local name is Mwitanzigye (killer of locusts). Sir Samuel Baker was the first European explorer to see Lake Albert. It was named Albert after Sir Albert who was the husband of Queen Victoria of England. Crude oil was discovered on the shores of lake Albert. Inland ports on Lake Albert.	Guided discovery Guided discussion Think pair share Brain storming			

<table>
<tr>
<td></td>
<td></td>
<td></td>
<td>Port Butiaba.
Port Ntoroko.
Port Wanseko.

Note: Port Butiaba is the main inland port on lake Albert.

<u>LAKE EDWARD.</u>
It is a rift valley lake formed by <u>faulting.</u>
It forms a natural boundary between Uganda and Democratic Republic of Congo.
It is connected to Lake George by Kazinga channel.
Kazinga channel has the biggest population of hippopotamuses in Uganda.
Kazinga channel is located in Queen Elizabeth national park.
<u>Henry Morton Stanley</u> was the first European explorer to see Lake Edward.

<u>LAKE GEORGE.</u>
It is the swallowest lake in Uganda formed by <u>faulting</u>.
It is located in Western Uganda.
It crossed by the Equator.
It is connected to Lake Edward by Kazinga channel.</td>
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<tr>
<td></td>
<td>14</td>
<td>✓ Describes crater lakes, lagoon lakes, manmade lakes, lava dammed lakes/volcanic lakes and oxbow lakes</td>
<td>
<table>
<tr><td><u>CRATER LAKES</u></td><td><u>Description.</u>
These are formed on top of volcanic mountains.

They are formed as a result of volcanicity.

Note:-Most crater lakes are found in south western Uganda because this region experienced volcanicity for a long time.</td><td><u>Examples.</u>
✞ Lake Katwe
✞ Lake Nyamunuka
✞ Lake Nyakasura
Note:-Lake Katwe is famous for salt production</td></tr>
<tr><td><u>LAGOON LAKES</u></td><td><u>Description.</u>

These are formed as a result of <u>marine deposition.</u></td><td><u>Examples</u>
• Lake Nabugabo.</td></tr>
</table>
</td>
<td></td>
<td>Guided discovery

Guided discussion

Think pair share

Brain storming</td>
<td></td>
<td>Describing lakes

Identifying types of lakes in Uganda</td>
<td>Study tours

vacations

Swimming

Fishing project</td>
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</table>

				They are formed when sand or mud separates the sea water/ part of a lake.						
				MAN MADE LAKES	Description They are formed as a result of man's activities/ dam construction. They are formed when man constructs a dam along rivers. They can also be formed as a result of digging valley dams in swamps. Most man-made lakes are usually small.	Examples ❖ Kabaka's lake				Games and sports
				LAVA DAMMED LAKES	Description They are formed when lava flows and blocks a flowing river.(as a result of volcanicity) Then water collects behind to form a river.	Examples Lake Bunyonyi (the deepest in Uganda) Lake Mutanda Lake Bulela				
				OX-BOW LAKES These are formed as a result of river deposition. They are mainly formed on meandering rivers in the lower stage of a river. ***A diagram showing the formation of ox-bow lakes.***						

RIVERS IN UGANDA	15	✓ Describes the stages of river development ✓ Describes river Nile ✓ Explains why some parts of river Nile are not navigable	RIVERS IN UGANDA A river is a mass of flowing water on the earth's surface. Many rivers in Uganda originate from high plateaus and highlands because such areas receive plenty of rainfall which provides water to the rivers. Rivers flow from areas of high altitude to areas of low altitude. Rivers continuously get their water from rainfall, melting snow or underground springs. **Examples of major rivers in Uganda** River Nile River Kafu River Mayanja River Kagera River Mpologoma River Achwa River Katonga River Sezibwa River Nkusi River Semliki River Manafwa **Types of rivers** Permanent rivers. These are rivers which flow throughout the year Examples of permanent rivers. River Nile River Kafu River Kagera River Sezibwa River Mpologoma River Achwa River Katonga Seasonal rivers. These are rivers that mainly flow during the wet season and dry up during the dry season. **Terms related to rivers.** A river source is a point where a river begins to flow. A river mouth is a point where a river ends its flow. A tributary is a small river that joins the main river. A distributary is a small river that branches away from the main river. A river confluence is a place where two or more rivers meet. An estuary is a wide part of a river as it ends its flow. A drainage basin is an area of land drained by a river, its tributaries and distributaries. A flood plain is a flat area near a river that often floods when the water level rises. A delta is the place where a river forms several streams as it ends its flow eg.The Nile delta.	Guided discovery Guided discussion Think pair share Brain storming		Describing rivers Identifying types of rivers in Uganda	Study tours vacations Swimming Fishing project Games and sports

Note:

Deltas are formed at the mouths of rivers, as a result of river deposition.

Deltas have rich fertile soil for crop growing.

They also attract tourists who bring in income.

A waterfall is a steep flow of a river.

Or.It is a point where a river flows from high altitude to a low altitude.

A Cataract is a series of waterfalls on a river.

Waterfalls attract tourists who bring in income, and also help in generation of hydroelectricity.

However, waterfalls may cause water accidents and also block water transport.

Location of major rivers on the map of Uganda.

Stages of River development

A river has three main stages/ courses.

These are;

The Youth stage/Upper course

The Old stage/ Lower stage (Senile)

The Mature stage/ Middle stage

Stage	Characteristics
The Upper course	• It has waterfalls and rapids • The river flows very fast. • It forms a V-shaped valley (gorge). • The river flows over a steep slope. • There is a lot of vertical erosion. Note: *This stage is suitable for generation of hydro electricity because it has many waterfalls.* *The upper stage is suitable for tourism because it has*

	many waterfalls which attract tourists.
The Middle stage	• The river flows gently. • It forms a U-shaped valley. • There is lateral erosion in this stage. It begins to form meanders.
The Lower stage	• The river flows slowly. • It forms meanders and ox-bow lakes. • It forms a delta or an estuary. • It forms alluvial plains. The river deposits its load forming flood plains.

<u>Features found along river valleys.</u>

Waterfalls

Gorges

Deltas

Ox-bow lakes

River Nile is the longest river in the world. Its local name is <u>Kiira.</u> It flows northwards because the north is on a low altitude.	Source	-Lake Victoria
	Mouth	-Mediterranean sea
	Countries drained	-Uganda -Sudan -Ethiopia -Egypt -South Sudan
John Hanning Speke was the first European explorer to see the source of river Nile.	Main tributaries	-River Achwa in Uganda
	Waterfalls	-Karuma falls -Murchison falls -Itanda falls
	Hydro electric power dams	-Nalubaale power dam -Kiira dam

			-Bujagali power dam -Karuma dam.					
			Nile valley countries	-Uganda,-South Sudan, -Ethiopia, -Sudan, -Egypt. NB:-*The Nile valley is the area drained by river Nile, its tributaries and distributaries.*				
			Sections of River Nile	-Victoria Nile (between lake Victoria and lake Albert) -Albert Nile (between lake Albert and Nimule) -White Nile (from Nimule northwards)				
				Reasons why some parts of River Nile are not navigable. Presence of waterfalls and rapids. Presence of floating vegetation/suds. Presence of dangerous aquatic animals. Some parts of river Nile are shallow and narrow.				
	16	✓ Describe river Katoga, Kagera, Kafu,Semlik and Achwa ✓ Explains activities carried out on and around lakes and rivers ✓ Explains the importance of lakes and rivers ✓ Illustrates the formation of convectional rainfall ✓ Explains problems	River Katonga. o It flows from lake George to Lake Victoria. River Kagera. o It forms a natural boundary between Uganda and Tanzania. o It flows from highlands in Burundi to lake Victoria in Uganda. River Kafu. o It flows from lake Albert to lake Kyoga. River Semliki. o It connects lake Edward to lake Albert. o It forms a natural boundary between Uganda and Democratic Republic of Congo. River Achwa. o It is the main tributary of River Nile in Uganda. o It joins river Nile near Nimule in South Sudan. **Activities carried out on and around lakes and rivers.** Fishing.	Guided discovery Guided discussion Think pair share		Describing rivers Identifying types of rivers in Uganda	Study tours vacations Swimming

		associated with lakes and rivers ✓ Explains problems facing lakes and rivers ✓ Explains why some parts of rivers in Ugandaare not navigable (not used for transport) ✓ Explains why lake shore and river banks are densely populated	Crop growing Tourism Transportation. Trading. Industrialisation. **<u>Importance of lakes and rivers</u>** They help in rain formation. They act as fishing grounds. Rivers help in generation of hydro electricity. They attract tourists who bring in income. They are homes for aquatic animals. They form natural boundaries between countries. They provide water for irrigation, domestic and industrial use. ***<u>A diagram showing the formation of Convectional rainfall.</u>*** **<u>Problems associated with lakes and rivers.</u>** Drowning Rivers and lakes hinder road and railway construction They harbour disease vectors Flooding They harbour dangerous aquatic animals. **<u>Problems facing lakes and rivers</u>** Silting which reduces the depth of water bodies. Dumping of wastes into water bodies. Overuse of water from lakes and rivers for irrigation. Over fishing. Drought which lowers the water level. **<u>Why some parts of rivers in Uganda are not navigable (not used for for transport)</u>** Some rivers have waterfalls and rapids. Some parts of rivers are shallow and narrow Some rivers have dangerous aquatic animals. Some rivers have floating vegetation. (suds) **<u>Reasons why lake shores and river banks are densely populated.</u>**	Brain storming			Fishing project Making electricity Games and sports

			They have fertile soils for crop growing. They receive reliable rainfall. They have a lot of employment opportunities.				
THE INFLUENCE OF PHYSICAL FEATURES	17	✓ Describes the influence of physical features on people ✓ Explains the factors which attract human settlement in plateaus and highland area ✓ Describes common economic activities carried out in highland areas ✓ Explains the reasons why there are few people living in higher parts of mountains	THE INFLUENCE OF PHYSICAL FEATURES a) <u>On People</u> The wind ward side of a mountain attracts very many people due to the presence of fertile soils for cultivation, and also receives plenty of rainfall. The lee ward side of a mountain has very few people because it receives very little rainfall. Plains favour pastoralism due to the presence of pasture. Road and railway transport is poor in highland areas due to the presence of steep slopes. Many people settle around lakes and rivers because such areas receive reliable rainfall and have fertile soils which support crop cultivation. <u>Factors which attract human settlement in;</u> i) <u>Plateau areas</u> Presence of pasture for animals. Agricultural mechanisation is easy in such areas. Construction of houses, roads, railways and industries is easy. Natural hazards like floods are not common in these areas. ii) <u>Highland areas</u> Presence of fertile soils. Availability of reliable rainfall. <u>Common economic activities carried out in highland areas.</u> Crop farming Mining Dairy farming <u>Reasons why there are very few people living in higher parts of mountains.</u> Transport is poor in higher parts of the mountains Highland areas experience very low temperatures which do not favour human settlement. b) <u>On Plants</u> ✞ Areas around lakes and rivers have many plants due to plenty of rainfall received.	Guided discovery Guided discussion Think pair share Brain storming		Describing physical features' influence on man	Study tours farming vacations Swimming Fishing project Games and sports

<table>
<tr><td></td><td></td><td></td><td>✞ Slopes of mountains have forests due to presence of fertile soils and reliable rainfall received.
✞ Mountain tops have few plants due to low temperatures and bare rocks. c) <u>On Animals.</u>
✞ Plains have plenty of grass which attracts wild animals.
✞ Water bodies favour aquatic animals.
✞ Pastoralism in common on plateau areas due to the presence of pasture for animals.
✞ There are many climbing animals eg.mountain gorillas in highlands due to the presence of many fruit trees.

d) <u>On Climate.</u>
✞ Mountains receive plenty of relief rainfall.
✞ The wind ward side of a mountain receives plenty of rainfall because it receives warm moist air.
✞ The lee ward side of the mountain receives very little rainfall because it receives dry winds.
✞ Mountainous areas experience very low temperatures because they lie at a higher altitude.
✞ Riftvalleys and low lands experience very high temperature because they are of low altitude.
✞ Lake shores receive plenty of convectional rainfall.</td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td>✓</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>TOPIC</td><td colspan="7">CLIMATE OF UGANDA</td></tr>
<tr><td>SUBTOPIC</td><td>SESSION</td><td></td><td>FACTS/CONTENT</td><td>METHODS</td><td>RELEVANCE</td><td>LESSON ACTIVITY</td><td>SCENARIO</td></tr>
<tr><td>INTRODUCTION TO WEATHER</td><td>18</td><td>✓ Gives the meaning of weather meteorology, weather forecasting
✓ Explains the importance of weather forecasting
✓ Explains the types, elements and conditions of weather
✓ Describes the elements/ factors of weather</td><td><u>Weather</u> is the state of the atmosphere of a place at a given time.
<u>Meteorology</u> is the scientific study of weather.
<u>A meteorologist</u> is a scientist who studies weather.
Elements of weather are measured and recorded at a weather station/ meteorological centre.
The biggest meteorological centre/ weather station in Uganda is Entebbe meteorological centre.
<u>Weather forecasting</u> is the telling of the expected future weather conditions of an area.
<u>Importance of weather forecasting.</u>
✞ It helps farmers to plan well their farm activities. e.g. Planting, harvesting etc.
✞ It helps travellers to prepare for their journeys.
✞ It helps pilots and sailors to avoid air and water accidents.
<u>Types and conditions of weather.</u>
<table><tr><td>Type of weather</td><td>Condition of weather</td></tr><tr><td>Rainy weather</td><td>Rainy</td></tr><tr><td>Windy weather</td><td>Windy</td></tr><tr><td>Cloudy weather</td><td>Cloudy</td></tr><tr><td>Sunny weather</td><td>Sunny</td></tr><tr><td>Foggy weather</td><td>Foggy</td></tr></table><u>Elements of weather / factors of weather</u>
✞ Rainfall
✞ Sunshine</td><td>Guided discovery

Guided discussion

Think pair share

Brain storming</td><td></td><td>Describing weather

Identifying types of weather</td><td>Academic clubs

vacations

Water harvesting</td></tr>
</table>

<table>
<tr>
<td></td>
<td></td>
<td></td>
<td>✞ Cloud cover
✞ Air pressure/atmospheric pressure
✞ Wind
✞ Temperature
✞ Humidity</td>
<td></td>
<td></td>
<td>Identifying elements of weather</td>
<td>Project making

Games and sports</td>
</tr>
<tr>
<td></td>
<td>19</td>
<td>✓ Gives the meaning of weather instruments
✓ Describes a wind vane/ weather cock, wind sock, an anemometer and a rain gauge
✓ Describes the importance of a barometer, sunshine recorder, hygrometer, six’s thermometer and a Stevenson screen
✓ Describes weather instruments kept in a Stevenson screen</td>
<td><u>WEATHER INSTRUMENTS</u>
These are tools used in measuring and recording the atmospheric conditions.
<table>
<tr><td>Weather instrument</td><td>Importance</td></tr>
<tr><td>Wind vane / a weather cock</td><td>✞ It shows the direction of wind.
✞ A wind vane is placed in an open place or on top of buildings to prevent wind obstruction.
Note:-An arrow of a wind vane usually points to the direction from which wind is blowing.</td></tr>
<tr><td>A Windsock</td><td>✞ It shows the strength of wind.
✞ It’s usually found at the airports, air fields and at chemical plants.
Note:-A wind sock is not among the instuments of weather found at a weather station.</td></tr>
</table>
<table>
<tr><td>An Anemometer</td><td>✞ It measures the speed of wind.
✞ It has cups which trap wind and rotate as wind blows into them.
✞ The speed of wind is measured in kilometres per hour (km/hr)</td></tr>
</table></td>
<td>Guided discovery

Guided discussion

Think pair share

Brain storming

demonstration</td>
<td></td>
<td>Describing weather instruments</td>
<td>Academic clubs

Water harvesting

Project making

Games and sports</td>
</tr>
</table>

			A Rain gauge	✞ It is used to measure the amount of rainfall received in an area. ✞ It is always placed at least 30cm above the ground to prevent splashes and flowing water on the ground from entering the rain gauge. ✞ It should be placed 15cm below the ground to prevent the evaporation of water collected and to make it firm. ✞ It is placed in an open flat place to prevent obstruction of rain drops from entering the funnel in order to take accurate measurements. <u>Importance of different parts of a rain gauge.</u> ✞ <u>The funnel</u>-it directs water into the bottle. ✞ <u>Water bottle</u>-it collects the amount of rain water received. ✞ <u>Measuring cylinder</u>-it is used for measuring the amount of water collected in the bottle. Note:-*Rainfall is measured in <u>millimetres</u> in order to know the depth of water into the soil.*					
			A Barometer	✞ It is used to measure air pressure. ✞ Air pressure is measured in millibars. ✞ Measuring air pressure enables experts to predict storms.					
			A Sunshine recorder	❖ It shows the duration of sunshine in a day.					

				✞ It is used to measure humidity. ✞ <u>Humidity</u> is the amount of water vapour in the atmosphere.					
			Hygrometer						
			Six's thermometer	❖ It measures the highest and lowest temperatures of the day.					
			A STEVENSON SCREEN.	○ This is a wooden box in which delicate weather instruments are kept at a weather station. ○ It protects weather instruments from rainfall and direct heat. ○ The Stevenson screen is made of louvres to allow free air circulation in the box. ○ It is painted white to enable it reflect sunlight. ○ It is raised 1.5m above the ground to keep it free from splash water. It is made of wood to prevent it from absorbing heat.					
			<u>Weather instruments kept in a Stevenson screen.</u> • Barometer • Six's thermometer (Maximum and thermometer). • Hygrometer <u>Note</u> *The above instruments are kept in a Stevenson screen in order to protect them from being destroyed since they are delicate.* *The Ministry of Water and Environment is responsible for monitoring weather and climate, and giving advice to the public on weather conditions in Uganda*						

	20	✓ Describe the terms related to climate ✓ Explains the main aspects/ factors of climate ✓ Describes the types climate experienced in Uganda ✓ Locate climatic regions on the map of Uganda ✓ Describes equatorial climate ✓ Describes the characteristics of equatorial climate ✓ Illustrates climatic table and graph of equatorial climate ✓ Describes areas in Uganda that experience equatorial climate ✓ Discusses districts the experience equatorial climate ✓ Discusses the economic activities carried out in equatorial climatic region ✓ Describes crops grown in equatorial climatic region	Climate is the average weather condition of a place recorded for a long time. Climatology is the scientific study of climate. A climatologist is a scientist who studies climate. Main aspects/ factors of climate. • Rainfall • Temperature. Types of climate experienced in Uganda. ✞ Equatorial climate ✞ Semi-desert climate ✞ Tropical climate ✞ Montane climate CLIMATIC REGIONS IN UGANDA. A climatic region is a large area that experiences the same / similar weather patterns. Climatic regions in Uganda. ✦ Equatorial climatic region ✦ Tropical climatic region/ Savannah climatic zone ▪ Semi-desert climatic region/ savannah climatic zone. ✦ Montane climatic region/ mountain climatic zone The Climatic Regions of Uganda THE EQUATORIAL CLIMATE It is described as hot and wet throughout the year. It is experienced in areas along the Equator Characteristics of Equatorial Climate. ✞ It is hot and wet throughout the year . ✞ It mainly receives convectional rainfall.(between 1500mm – 2000mm) ❖ Rainfall is accompanied by lightning and thunderstorms. ✞ It has no dry month. ✞ Temperatures are high especially during equinox . The sun overheads the Equator twice a year on 21st March and 23rd September. ✞ It has two heavy maximum rainfall seasons (double maxima). A climatic table showing Equatorial type of climate. A climatic graph showing Equatorial type of climate	Guided discovery Guided discussion Think pair share Brain storming		Describing climate Identifying types of climate Identifying climatic zones	Academic clubs Weather forecastin g Water harvesting Project making Games and sports

			➢ Note:-*The increase in temperatures causes increase in the amount of rainfall received in the Equatorial climatic region.* <u>Areas in Uganda that experience Equatorial climate.</u> ✞ Shores of Lake Victoria ✞ The South Western part of Uganda. <u>Districts that experience Equatorial climate.</u> ✞ Wakiso ❖ Buikwe ❖ Kalangala ✞ Jinja ❖ Mukono ❖ Mpigi <u>Economic activities carried out in Equatorial climatic region</u> • Crop growing • Tourism • Lumbering <u>Crops grown in Equatorial climatic region.</u> • Oil palm • Coffee • Cocoa • Bananas • Rubber Note:-*The above crops are commonly grown because they require plenty of rainfall for them to grow well.*				
CLIMATIC REGION/ ZONE OF UGANDA		✓					
	21	✓ Describes tropical climate/savannah climate ✓ Discuses districts that experience tropical climate in Uganda ✓ Discusses the characteristics of tropical climate ✓ Illustrates the climatic table and graph of tropical climate ✓ Describes the economic activities commonly carried out in tropical climatic region	<u>TROPICAL CLIMATE / SAVANNAH CLIMATE</u> Tropical climate is described as hot and wet. Tropical type of climate is experienced in the largest part of Uganda. This is because Uganda lies between the tropics (with in the tropical region). <u>Districts that experience Tropical climate in Uganda.</u> ✞ Mubende. ✞ Soroti ✞ Kibale. ✞ Gulu ✞ Mbarara ✞ Luwero. ✞ Kitgum ❖ Sembabule. <u>Characteristics of Tropical climate</u> ✞ Rainfall is mainly received when the sun is overhead the equator. ✞ It has two wet seasons and two dry seasons. ✞ Rainfall decreases as one moves far from the equator. ❖ Tropical type of climate is hot and wet. <u>The climatic table showing Tropical type of climate.</u> <u>A climatic graph showing Tropical type of climate</u> <u>Economic activities commonly carried out in Tropical climatic region</u> • Crop growing • Bee keeping • Pastoralism • Charcoal making. • Tourism • Lumbering <u>Common crops grown in Tropical climatic region</u> • Maize • Sorgum	Guided discovery Guided discussion Think pair share Brain storming		Describing tropical climate Identifying economic activities in	Academic clubs Rain formation Water harvesting

		✓ Discuses common crops grown in tropical climatic region	• Sorghum • Cotton • Simsim • Ground nuts • Beans • Fruits like; jack fruits, oranges, mangoes, guavas			tropical climatic zones	Project making
	22	✓ Describes semi desert/ semi-arid climate ✓ Discusses the districts in Uganda that experiences semi desert climate ✓ Describes the characteristics of semi desert climate ✓ Illustrates the climatic table and climatic graph of semi desert climate ✓ Describes montane/ mountain climate ✓ Describes the areas that experience montane climate ✓ Discusses the districts in Uganda that experience montane climate in Uganda ✓ Describes the characteristics of montane climate ✓ Describes the economic activities commonly	<u>SEMI-DESERT CLIMATE/ SEMI-ARID CLIMATE.</u> It is described as hot and dry. It is experienced in North Eastern part of Uganda. <u>Districts in Uganda that experience Semi-Desertclimate.</u> ✞ Moroto ❖ Kaabong ✞ Kotido ❖ Nakapiripirit ✞ Abim <u>Characteristics of Semi-desert climate.</u> ✞ It is hot and dry. ✞ There is low humidity. ✞ It has very hot days and cold nights. ✞ Skies are clear with little cloud cover. ✞ It receives low and unreliable rainfall. <u>A Climatic table showing Semi-desert type of climate.</u> <u>A climatic graph showing Semi-desert type of climate</u> <u>Economic activities carried out in Semi desert climate.</u> ✞ Pastoralism. ✞ Crop cultivation. ✞ Tourism. <u>MONTANE/MOUNTAIN CLIMATE.</u> Montane climate is experienced in highland areas. Temperature reduces with the increase in altitude. Temperature reduces by 1°C for every 100 metres above the sea level. The wind ward side receives plenty of rainfall because it receives warm moist air. The lee ward side receives very little rainfall because it receives dry winds. <u>Some areas that experience Montane climate</u>. • Areas around mountain Rwenzori • Areas around mountain Mufumbiro • Areas around mountain Elgon <u>Districts in Uganda that experience Montane climate.</u> ✞ Kasese ❖ Mbale ❖ Kabale	Guided discovery Guided discussion Think pair share Brain storming		Describing semi-desert climate Identifying economic activities in semi-arid climatic zones	Academic clubs Rain formation Water harvesting Project making Dressing code

		carried out in montane climate	✞ Kanungu ❖ Sironko ❖ Kisoro ✞ Bundibugyo ❖ Bududa <u>Characteristics of Montane climate.</u> ❖ It has cool temperature. ✞ Relief rainfall is mostly received. ✞ The wind ward side receives more rainfall than the lee ward side. <u>Economic activities commonly carried out in Montane climate.</u> ✞ Tourism. ❖ Lumbering ✞ Crop growing.				
		✓					
FACTORS THA INFLUENCE OR AFFECT THE CLIMATE OF UGANDA	23	✓ Describes the factors that influence the climate of Uganda ✓ Explains how the above factors influence the climate of Uganda ✓ Discuses human activities that affect the climate of an areas positively ✓ Give the meaning of a breeze ✓ Describes the types of breezes	**<u>FACTORS THAT INFLUENCE CLIMATE</u>** ✞ Altitude. ✞ Wind movement. ✞ Nearness to water bodies/Drainage. ✞ Human activities ✞ Latitude. ✞ Vegetation cover. (i) <u>Altitude</u> <u>Altitude</u> is the height above the sea level. ✞ Temperature reduces with the increase in altitude. The lower parts of a mountain are warmer and have higher temperatures than the higher parts of the mountain. The higher the altitude, the cooler the temperature,and the lower the altitude, the hotter the temperature eg. Mbale is cooler than Kampala because Mbale is on a higher altitude than Kampala. (ii) <u>Vegetation</u> ✞ Places with thick vegetation usually receive plenty of rainfall while those that have scanty vegetation usually receive very little rainfall. Note:-*Vegetation modifies climate of an area by helping in the formation of convectional rainfall through transpiration.* (iii) <u>Human activities</u> Some human activities affect climate positively while others affect climate of an area negatively. <u>Human activities that affect climate of an area positively.</u> • Afforestation • Reafforestation • Agro-forestry	Guided discovery Guided discussion Think pair share Brain storming		Describing factors that influence climate climate	Academic clubs Rain formation Water harvesting Project making

			Human activities that affect climate of an area negatively. • Deforestation • overgrazing • Bush burning • Swamp drainage • Industrialisation (iv) Latitude. Latitude is the distance in degrees North or South of the equator. Areas which are near the Equator are hotter than those which are far away from the equator. Low latitude areas are areas which are near the equator, and are usually very hot. High latitude areas are areas which are far away from the Equator and are usually cool. Areas which are far away from the Equator are usually very cool because they receive slanting sun rays which travel a very long distance to the earth. Note; *Most parts of Uganda are always hot because the sun is overhead Uganda throughout the year.* *The Revolution of the earth (in 365days) causes changes in seasons. ie. wet season and dry season.* *Tropical region is the area of land lying between the Tropic of cancer and Tropic of capricorn.* *Equinox is the time of the year when the sun is overhead the equator.* *The days and nights are always equal whenever the sun is overhead the equator.* *Very high temperatures are experienced around equinoxes.* *Solstice is the time of the year when the sun is overhead the tropics.* A diagram showing the time when the sun is overhead major lines of latitude. (v) Nearness to water bodies (Drainage) Areas near large water bodies usually receive more rainfall than those which are far away from large water bodies. Areas near large water bodies usually receive convectional rainfall. Winds that blow over large water bodies pick a lot of moisture which rises and condenses to form convectional rainfall. Note*: -Water bodies modify climate by helping in the formation of convectional rainfall through evaporation.* Characteristics of convectional rainfall • It is mostly received in the afternoon. • It is usually accompanied by lightning and thunderstorms. • It is mostly received in areas that experience a lot of evaporation and transpiration.				

			(vi) <u>Prevailing winds</u>. Areas that receive warm moist air receive plenty of rainfall while those that receive dry winds usually receive plenty of rainfall. <u>BREEZES.</u> <u>A breeze</u> is the movement of air from a cool region to a warm region. <u>Types of breezes.</u> • Sea breeze • Land breeze Sea breeze This is the movement of cool air from the sea towards land. It takes place during the <u>day.</u> During the day, the land is heated faster than water bodies. The warm air on land rises and cold air from the sea moves towards land to replace it. NB:-*Sea breezes help in formation of convectional rainfall.* Land breeze This is the movement of cold air from land towards the sea. It takes place during the <u>night.</u> At night, the land cools faster than the sea. The warm air from the sea where pressure is low rises and the cooler air blows from land towards the sea to replace it.				
THE INLUENCE OF CLIMATE ON HUMAN ACTIVITIES	24	✓ Describes the economic activities commonly carried out in montane climate ✓ Describes the influence of montane climate on people ✓ Explains the problems faced by people living in montane climatic region	**<u>The influence of Equatorial and Tropical climate on human activities.</u>** Lumbering is done in the tropical rain forests. Tourism is favoured by the plenty of animals. Perennial crops are usually grown due to plenty of rainfall receive eg.coffee,oil palm,bananas,tea etc. People wear thick clothes to provide warmth during rainy and windy weather. People keep a variety of animals due to presence of pasture for animals. Houses are built with slanting roofs to allow easy flow of rain water off the roofs. **<u>Problems faced by people living in Equatorial and Tropical regions.</u>** Floods Attacks from wild animals Poor ground transport. Presence of many disease vectors like tsetse flies, mosquitoes etc. **<u>Possible solutions to the problems faced by people living equatorial and tropical</u> <u>climatic region</u>**	Guided discovery Guided discussion Think pair share		Identifying the influence of climate on man's activities	Farming Tourism Fishing studying

		✓ Describes the possible solutions to the problems facing people in montane climatic region	By building houses with slanting roofs. By wearing rain coats and gum boots to safeguard against rainy weather. By creating more wildlife conservation areas. c**) <u>Montane climate</u>** **<u>Economic activities commonly carried out in Montane climate</u>**. Tourism. Crop growing. Lumbering **<u>The influence of the montane climate on people.</u>** People build houses with slanting roofs. People rear donkeys for use as means of transport. People wear thick clothes to protect their bodies from much coldness. Thick forests promote lumbering in this climatic region. People grow perennial crops due to plenty of rainfall received. **<u>Problems faced by people living in Montane climatic region.</u>** Poor transport network. Severe soil erosion Landslides. Much coldness **<u>Possible solutions to the problems faced by people living in montane climatic region</u>** By building winding roads. By keeping donkeys for transport. By wearing thick clothes to overcome much coldness. By planting trees to reduce soil erosion and landslides. NB: *Plant roots control landslides in highland areas by holding the soil particles firmly.*	Brain storming			

FARMING IN UGANDA		✓ Gives the meaning of farming ✓ Explains the types of farming ✓ Describes the systems of crop growing ✓ Explains the meaning of subsistence farming ✓ Discusses the advantages and disadvantages of subsistence farming ✓ Gives the meaning of mixed farming Discusses the advantages and disadvantages of mixed farming	**FARMING IN UGANDA.** **Farming** is the growing of crops and rearing of animals. It is also called agriculture. Farming is the main economic activity carried out by most people in Uganda. **Types of farming.** Arable farming. Livestock farming. **Systems of crop farming.** Subsistence farming. Plantation farming. **SUBSISTENCE FARMING.** This is the growing of crops and rearing of animals mainly for home use and the surplus is sold. It is the commonest system of farming practiced in Uganda. This is because subsistence farming is very cheap to practice. In this system of crop growing, annual crops are mainly grown eg. Beans, maize, cassava, peas, irish potatoes etc. **Advantages of subsistence farming.** It is cheap to practice. It requires simple tools. It does not require a large piece of land. It requires less labour force. It promotes food security. It does not require a large piece of land. **Disadvantages of subsistence farming.** A farmer earns less income. It leads to low productivity. It does not encourage economic development. **MIXED FARMING.** **Mixed farming** is the growing of crops and rearing of animals on the same piece of land. **Advantages of mixed farming.** Farmers gets double income. The farmer can get manure for his crops from animals.			Describing farming Identifying systems of farming Describing the systems of farming	School garden Academic clubs Project work making

			Animals can easily depend on crops for food. A farmer gets a balanced diet. <u>Disadvantages of mixed farming.</u> It is expensive to start and manage. It requires skilled labour. Animals can easily escape and destroy the crops. It requires a large piece of land.				
	25	✓ Gives the meaning of plantation farming ✓ Describes the advantages and disadvantages of plantation farming ✓ Describes the examples of plantation crops Describe the districts where plantation crops are commonly grown	**<u>PLANTATION FARMING.</u>** This is the growing of one perennial crop on a large scale. Plantation farming is practised by very few people in Uganda because it is very expensive to practice. Under plantation farming, perennial crops are mostly grown eg. coffee, tea, cocoa, oil palm etc. **<u>Advantages of plantation farming</u>**. It is a source of employment to people. It is a source of raw materials for industries. It is a source of income to the government through taxing plantation owners. It promotes the development of roads. Note: *Agro-based industries are industries which use agricultural produce as their raw materials.* **<u>Disadvantages of plantation farming.</u>** It is very expensive to carry out. It requires a large piece of land. Crops can easily be affected by drought. Pests and diseases can easily spread from one crop to another. It leads to soil exhaustion. **<u>Examples of plantation crops.</u>** Tea. Coffee. Cloves. Sugarcane Cocoa. Cotton. Oil palm Sisal. **<u>TEA</u>** Tea is a beverage crop. **<u>Examples of plantations for tea.</u>** Kasaku tea plantation. Igara tea plantation.	Guided discovery Guided discussion Think pair share Brain storming		Describing plantation farming Describing examples of plantation farming	School garden Academic clubs Project work making Study trips

			Nakigala tea plantation. **<u>Tea growing districts.</u>** Mukono. Rukungiri. Kabalore. Bushenyi. Mubende. **<u>Conditions for growing tea.</u>** Warm temperatures. Reliable rainfall. Acidic soils.				
	26	✓ Describes coffee growing in Uganda ✓ Discusses the types of coffee ✓ Explains the advantages of growing clonal coffee over other types of coffee Describes the factors that favor the growing of arabica coffee on the slopes of mount Elgon	**<u>COFFEE.</u>** It is a beverage crop. It contains caffeine. **<u>Types of coffee.</u>** Arabica coffee (grown in highlands). Robusta coffee (grown in low lands). Clonal coffee (improved robusta). **<u>Advantages of growing clonal coffee over other types of coffee.</u>** It is more resistant to diseases than other types. It matures faster than other types. It gives more yields than other types. **<u>Factors tha favour the growing of arabica coffee on the slopes of mountain Elgon.</u>** Presence of deep volcanic fertile soils. Presence of heavy rainfall received. *Coffee is harvested by hand picking of berries.* *Arabica coffee is mainly grown by the Bagishu on the slopes of mountain Elgon and the Bakonzo on the slopes of mountain Rwenzori.*	Guided discovery Guided discussion Think pair share Brain storming		Describing coffee growing in Uganda	School garden Academic clubs Factory visits Project work making

							Study trips
		✓ Describes sugar cane growing in Uganda ✓ Describes wheat growing in Uganda ✓ Describes cotton growing in Uganda	**SUGARCANE** Sugarcane stems are crushed to get juice which makes sugar crystals. **Sugarcane plantations in Uganda.** Kakira sugarcane plantation in Jinja. Lugazi sugarcane plantation in Buikwe. Kinyara sugar works in Masindi. Sango bay in Rakai. **WHEAT.** It is a cereal crop which requires low temperature and heavy rainfall. Grain provide flour used in baking. **Districts that grow wheat.** Kapchorwa. Kisoro. Bundibugyo. **COTTON.** It is a fibre crop used in textile industries. It is grown using seeds and harvested by hand picking. It is taken to ginneries to remove seeds and remain with lint. The lint is processed into threads by spinning. **Kenneth Borup** was a missionary who introduced the fast growing cotton seeds in Uganda in 1903. **Sir Hesketh Bell** encouraged cotton growing in uganda. **Why the British colonialists encouraged cash crop growing in uganda.** They wanted to get raw materials for their home industries. They wanted the natives to get income and pay taxes.			Describing sugarcane growing in Uganda	School garden Academic clubs Factory visits Project work making Study trips

			Note: *The introduction of synthetic fibres eg.Nylon has reduced the market for cotton hence leading to reduction in cotton production.* Textile industries areindustries that use cotton as their raw material eg. Nyanza Textile Imdustries Limited (NYTIL) **<u>Cotton growing districts in Uganda.</u>** Kasese. Iganga. Kamuli. **<u>Uses of cotton.</u>** It is used for making threads. It is used for making clothes. It is used making cotton wool. Cotton seeds are used to make animal feeds. **<u>Problems faced by cotton growers</u>.** Cotton pests and diseases. Harsh climate changes. Fluctuation of cotton prices. Competition from cotton growers.				
	27	✓ Describes tobacco growing in Uganda ✓ Describes oil palm growing in Uganda ✓ Explain the types of crops ✓ Describes traditional cash crops	**<u>TOBACCO</u>** It is grown in West Nile It is used to make ciragettes British America Tobacco (BAT) is the body responsible for marketing and processing tobacco. The introduction of the tobacco control bill has led to the reduction in tobacco production in West Nile. **<u>OIL PALM</u>**. Oil palm is mainly grown in Kalangala by BIDCO oil company. Oil palm is used to get palm oil. **<u>Products got from palm oil</u>.** Palm Wine Soap Cooking Oil Margarine Candles **<u>Factors that favour oil palm growing.</u>**	Guided discovery Guided discussion Think pair share Brain storming		Describing tobacco growing in Uganda	School garden Academic clubs Factory visits

			Presence of heavy rainfall. Presence of high temperature. Well drained fertile soils. **Note:** *Harvested oil palm nuts are transported to Jinja for processing and Bidco cooking oil, Bidco washing soap and other products are obtained.* **<u>TYPES OF CROPS.</u>** Traditional cash crops. Non-traditional cash crops. **(i) <u>TRADITIONAL CASH CROPS</u>.** Traditional cash crops are crops which were originally grown for sale. **<u>Examples of traditional cash crops</u>.** Coffee — Cotton Tobacco — Tea Cocoa — Oil palm Sisal — Rubber **<u>Why people are encouraged to grow traditional cash crops.</u>** To promote export trade. To get raw materials for industries. **<u>Products obtained/ made from different cash crops</u>**				Project work making Study trips

Crop	Product(s)
Cotton	Clothes, cotton wool, threads
Pyrethrum	Insecticides
Oil palm	Palm oil used for making cooking oil, soap, margarine etc.
Tea	Tea leaves
Tobacco	Cigarettes
Sun flower	Cooking oil
Sisal	Ropes, strings, sacks
coffee	Coffee powder, gun powder.
Maize	Maize flour
Rubber	Latex for making gloves,balls, shoe soles, car tyres, erasers etc

	28	✓ Gives the meaning of non-traditional cash crops ✓ Explains why the government is encouraging people to grow more non-traditional cash crops ✓ Describes the examples of non-traditional cash crops ✓ Describes perennial and annual crops	**(i) NON-TRADITIONAL CASH CROPS.** **Non-traditional cash crops** are crops which were originally grown for food but can now be sold. **Why the government is encouraging people to grow more non-traditiional cash crops.** They have a large market. To promote food security in the country. **Examples of Non-traditional cash crops.** Beans, Banana, Ground nuts Maize, Yams, Soya beans Cassava, Rice, Vegetables like cabbage **PERENNIAL CROPS.** These are crops which take a long time to mature and are harvested several times. Most of the traditional cash crops are perennial. **Examples of perennial crops.** Coffee Cocoa Sisal Tea Mango plants Rubber Oil palm **ANNUAL CROPS.** These are crops which take a short time to mature and are harvested once. **Examples of annual crops.** Maize, Cotton, Sorghum Beans, Tobacco, Cassava Rice, Millet, Peas	Guided discovery Guided discussion Think pair share Brain storming		Describing non traditional cash crops growing in Uganda Identifying perennial crops Identifying annual crops	School garden Academic clubs Factory visits Project work making Study trips

IRRIGATION FARMING IN UGANDA	29	✓ Explains terms related to irrigation ✓ Describes irrigation schemes in Uganda ✓ Explains common methods of irrigation Describes the advantages and disadvantages of irrigation farming	**<u>IRRIGATION FARMING IN UGANDA.</u>** **<u>Irrigation</u>** is the artificial supply of water on land to support plant growth. **<u>Irrigation farming</u>** is the system of crop growing in which land is supplied with water by human means to support crop growth. **<u>Irrigation scheme</u>** is an area of land which is supplied with water by human means to support crop growth. Irrigation schemes are mostly set up in areas that do not receive reliable rainfall (dry areas). These areas must be having reliable permanent sources of water. Irrigation farming is not well developed in North Eastern uganda because the area has few reliable permanent sources of water. **<u>Irrigation schemes in Uganda.</u>** Doho irrigation scheme for rice Tilda (Kibimba) irrigation scheme. Kiige irrigation scheme for citrus fruits. Ongom irrigation scheme in Lira for citrus fruits. Olweny swamp irrigation scheme in Lira for rice Nsimbe and Rosebud irrigation schemes for flowers **<u>Common methods of Irrigation.</u>** Gravity flow method Overhead sprinkler method **<u>Advantages of Irrigation farming.</u>**	Guided discovery Guided discussion Think pair share Brain storming		Describing irrigation farming in Uganda Identifying irrigation schemes in Uganda	School garden Academic clubs Factory visits Project work making Watering crops Study trips

Irrigation scheme	Source of water	District / country	Crops grown
Doho	R. Manafwa	Butaleja	Rice.
Mubuku	R. Mubuku And R. Sebwe	Kasese	Vegetables, maize
Tilda (Kibimba)	R. Mpologoma	Bugiri	Rice
Sango Bay	L. Victoria	Rakai	Sugarcanes and Maize
Kakira	L. Victoria	Jinja	Sugarcanes
Lugazi	L. Victoria	Buikwe	Sugarcanes
Pabo	R. Achwa	Kitgum	Rice
Kiige	L. Nabigaga	Kamuli	Citrus fruits
Agoro	R. Agago	Lamwo	Tomatoes, Okra, Maize

			Crops are grown at any time of the year. It promotes food security in the country. Irrigation farming helps to put idle dry land into use. Crops grow well without being affected by sunshine. **<u>Disadvantages of Irrigation farming.</u>** It is very expensive to practice. It is not possible where there is no reliable source of water. It leads to soil leaching. **Note:** *Leaching is the sinking of soil nutrients to deeper layers where plant roots can not reach.*				
	30	✓ Describes ways of caring for crops in the garden ✓ Discuses the factors that have promoted farming in Uganda ✓ Explains the importance of farming ✓ Describes the problems faced by farmers ✓ Explains the possible solutions to the problems affecting agricultural development in Uganda	**<u>Ways of caring for crops in the garden.</u>** By weeding them. By thinning. By pruning them. By watering the crops. **<u>Factors that have promoted farming in Uganda</u>** Presence of fertile soils. Presence of reliable rainfall. Availability of ready market for agricultural produce. Improved transport and communication network. Political stability in an area. Availability of ready market for agricultural produce. Presence of large labour force. **<u>Importance of farming.</u>** It is a source of food to people. It is a source of raw materials for industries. It has promoted the development of roads and railways. It is a source of employment to people. It is a source of revenue to the government. **<u>Problems faced by farmers.</u>** Shortage of funds/ limited capital. Prolonged drought. Outbreak of crop pests and diseases. Limited market for farm produce. Poor transport network. Political instability in some parts of the country. Shortage of land for crop growing in some areas eg in towns.	Guided discovery Guided discussion Think pair share Brain storming		Identifying ways of caring for crops Stating the importance of farming	School garden Academic clubs Factory visits Project work making Study trips

			Changes in prices/unstable prices for agricultural produce. Rural-urban migration which reduces labour force on farms in rural areas. **<u>Possible solutions to the problems affecting agricultural development in Uganda.</u>** By giving soft loans to farmers. By spraying the crops with insecticides. By teaching farmers better farming methods. The government should set minimum prices for crop products. By building better roads in the country. By encouraging the use of irrigation farming. **Note:** *The government introduced developmental programmes such as National Agricultural Advisory Services (NAADs) and Operation wealth creation to help farmers to improve their lives.* *NAADs is working under the Ministry of Agriculture, Animal industry and Fisheries.* *The government also set up the National Agricultural Research Organisation (NARO) to carry out research on how to improve farming activities.*				
LIVESTOCK FARMING IN UGANDA	31	✓ Gives the meaning of pastoralism ✓ Describes the systems of livestock farming ✓ Describes dairy farming ✓ Describes ranching ✓ Gives the meaning of nomadic pastoralism ✓ Describes the examples of pastoral tribes in Uganda ✓ Discusses the districts where nomadic pastoralism is practiced ✓ Discusses reasons why	**<u>LIVESTOCK FARMING IN UGANDA.</u>** **<u>Pastoralism</u>** is the rearing of cattle on a large scale. Livestock farming is also called **pastoralism.** **<u>Systems of livestock farming</u>.** Dairy farming. Nomadic pastoralism. Cattle ranching. **<u>DAIRY FARMING.</u>** **<u>Dairy farming</u>** is the rearing of cattle on a large scale mainly for milk production. Cows are mainly reared in this system of pastoralism. It is commonly carriedout in the cool mountainous areas. Dairy farms in Uganda commonly found in Kisoro and Kabale. **<u>Products from a dairy farm.</u>** Milk. Meat (beef). Hooves Hides. **<u>Milk products.</u>** Cheese. Yoghurt.	Guided discovery Guided discussion Think pair share Brain storming		Describing livestock farming in Uganda Describing dairy farming	School garden Academic clubs Factory visits Project work making

		pastoralists move from one place to another ✓ Describes the problems faced by the pastoralists in Uganda and their possible solutions	Ice cream. Ghee. Butter. **RANCHING.** Ranching is the rearing of cattle on a large scale mainly for beef production. Bulls are mainly reared in this system of livestock farming. **Products obtained from a ranch.** Beef. Hooves. Horn tips. **Examples of ranches in Uganda.** Kisozi ranch (Gomba district). Singo ranch Zziwa ranch Kabula ranch in Sembabule. Nyabushozi in Kiruhura. Buruli in Nakasongola. NOMADIC PASTORALISM. **Nomadic pastoralism** is the system of animal rearing where farmers move with their animals from one place to another in search for pasture and water. It is mainly carried out in semi-arid areas. Nomadism is commonly practiced by Karimojongs in the North Eastern Uganda. **Examples of pastoral tribes in Uganda.** Karimojongs. Bahima. **Districts where nomadic pastoralism is practiced**. Moroto district. Kotido district. Nakapiripirit district. **Reasons why pastoralists move from one place to another.** To look for pasture for their animals. To look for water for their animals. To break the lifecycle of disease vectors. **Problems faced by the pastoralists in Uganda.**				Study trips

			Cattle rustling. Poor breeds of cattle. Shortage of water for animals. Shortage of pasture. Outbreak of animal diseases. Poor veterinary services **<u>Possible solutions to the problems facing pastoralists in Uganda</u>**. Disarming cattle rustlers. Constructing valley dams to provide water for animals. Growing fodder crops to provide food to animals. By extending veterinary services to the pastoral tribes. By encouraging pastoralists to keep a limited number of animals.				
TOPIC	NATURAL RESOURCES IN UGANDA.						
SUB-TOPIC	SESSION	COMPETENCES	FACTS/CONTENT	METHODS	RELEVANCE	LESSON ACTIVITY	SCENARIO
INTRODUCTION TO RESOURCES IN UGANDA	32	✓ Gives the meaning of a resource. ✓ Gives the examples of resources ✓ Describes the examples of resources ✓ Describes the examples of natural resources ✓ Describes land as a resource Discusses land degradation	**<u>INTRODUCTION TO RESOURCES IN UGANDA</u>**. **<u>A Resource</u>** is any component of the environment that can be used to satisfy man's needs. Resources are classified into natural resources and man-made resources. **<u>Examples of resources.</u>** Land Wind/ air Water Minerals Animals People Cars Sunshine Computers Telephones. Buildings **<u>Natural resources</u>** are things in the environment that exist on their own and can be used to satisfy man's needs.	Guided discovery Guided discussion Think pair share Brain storming		Describing resources Identifying examples of resources Describing types of resources	Academic clubs Factory visits Project work making Farming Watering crops

			<u>Types of natural resources.</u> Renewable resources. Non-renewable resources. **(i)** **<u>Renewable resources</u>** are things which are used to satisfy man's needs that cannot get exhausted when used. These resources cannot get used up whenever people use them. Renewable resources can be replaced naturally after use. Renewable resources are also referred to as **<u>inexhaustible resources</u>**. **<u>Examples of renewable resources.</u>** Land Water Plants/ vegetation People Wind Animals Sunshine **(ii)** **<u>Non-renewable resources</u>** are things which are used to satisfy man's needs that can get exhausted when used. These resources can get used up when people use them. Non-renewable resources cannot be replaced naturally after use. Non-renewable resources are also referred to as **<u>exhaustible resources.</u>**				Study trips

			<u>Examples of non-renewable resources.</u> Fossil fuels Minerals eg.petroleum, sand, gold etc. Note: *Man made resources are things made by man and are used to satisfy man's needs.eg cars, aeroplanes, telephones, computers, trains etc.* **<u>LAND AS A RESOURCE.</u>** Land is the surface of the earth's surface that is not covered by water. Land is the most important resource because; Most resources are found on land. eg.plants, animals, minerals etc. Most economic activities are carried out on land. **<u>Uses of land</u>**. Land is used for crop growing. Land is used for construction of houses, markets, roads, factories et Land is used for grazing animals. Land is used as a mining ground. Land is sold to get money. **<u>LAND DEGREDATION.</u>** **<u>Land degredation</u>** is the misuse of land lowering its quality and productivity. **<u>Ways people degrade land.</u>**				

			Through deforestation. Through brick making. Through dumping wastes on land. Through bush burning. Through over grazing. **<u>Land conservation</u>**. This is the protection of land from being destroyed. **<u>Ways of conserving land.</u>** Through afforestation / planting trees. Through contour ploughing. Through bush fallowing. By terracing. Proper disposal of waste materials. By applying fertilizers. Through agro-forestry.				
	33	✓ Describes water as a resource ✓ Discusses the importance of water sources ✓ Discusses how man mis uses water as a resource ✓ Discusses the dangers of water pollution ✓ Describes ways of conserving water sources	**<u>WATER RESOURCES.</u>** These are important things got from water sources. Sources of water resources include; Rivers Swamps Lakes Ponds Streams	Guided discovery Guided discussion Think pair share		Describing water resources	Academic clubs Factory visits Project work making

		✓ Gives the meaning of fishing ✓ Discusses the fish species caught in Uganda Describes modern and traditional methods of fishing.	**<u>Importance of water sources.</u>** They promote fishing. They are a source of raw materials for industries. They promote tourism. They are used as a medium of water transport. They create job opportunities to people. Rivers help in generation of hydro electricity. **<u>How man misuses water sources.</u>** Through dumping garbage in water sources. Through over fishing. Through over use of water from water sources. **<u>Dangers of water pollution.</u>** It leads to death of fish in water bodies. It leads to outbreak o water borne diseases. It reduces the depth of water bodies. **<u>Ways of conserving water sources.</u>** By fencing wells. By planting trees near water sources. By using better fishing methods. By bathing in water sources. By defecating in water bodies. By urinating in water sources.	Brain storming			Farming Watering crops Study trips Fish ponds

			By dredging wells. **<u>FISHING IN UGANDA.</u>** **<u>Fishing</u>** is the extraction of fish from water bodies. Fishing in Uganda is carried out in rivers, lakes, ponds and streams. Lake Victoria is the largest inland fishing ground in Uganda.. Nile perch is the largest fish species caught in Uganda while Tilapia is the commonest type caught. **<u>Other fish species caught in Uganda.</u>** Mud fish (mostly got from swamps) — Sprat fish Lung fish — Sardine fish Cat fish — Eel fish Fish caught in Uganda is sold locally to the consumers while some is exported to other countries. **Traditional fishing methods used in Uganda include;** Use of baskets, hooks, spears and using bows and arrows. **Modern methods of fishing** such as, gill net method, trawling, line fishing method and drifting are also used. Fish is preserved locally by smoking it, sun drying it and salting it. Refrigeration and tanning/canning are the commonly used modern methods of fish preservation in Uganda.			Describing fishing in Uganda	
	34	✓ Describes the importance of fishing industry. ✓ Discusses the problems facing fishing industry.	**<u>Importance of the fishing industry</u>.** It is a source of employment opportunities to people. Exported fish earns foreign exchange to the government.	Guided discovery		Identifying importance of fishing in Uganda	Academic clubs

		✓ Describes the dangers of water hyacinth ✓ Discusses the ways of controlling of controlling water hyacinth. Describes the possible solutions to the problems facing fishing industry.	It is a source of food to people. Fish is used as a raw material in fish processing industries. Fish is a source of income when sold. Fish is used for making drugs e.g. Cod liver oil. It promotes the development of ports and landing sites. **<u>Problems facing the Fishing industry in Uganda.</u>** Indiscriminate fishing (catching of young fish) which leads to extinction of fish species. The use of undersized fishing nets leads to catching of young fish. Water pollution which kills fish. Presence of water hyacinth. Limited capital. Poor fishing methods eg use of poison. Poor storage facilities. Presence of fish predators like crocodiles. Fluctuation/changes of fish prices. Cultural beliefs which discourage some people from eating fish. Limited local market for fish. This is due to high levels of poverty among the people. **<u>Dangers of water hyacinth.</u>** It suffocates fish. It breaks fishing nets. It hinders water transport. It harbours some disease vectors. It blocks fishing grounds. **<u>Ways of controlling water hyacinth.</u>**	Guided discussion Think pair share Brain storming			Factory visits Project work making Farming Watering crops Study trips

			By spraying it with chemicals. By using manual methods to remove it. By using machines to remove it. By introducing beetles to feed on it. **<u>Possible solutions to the problems facing the Fishing industry.</u>** By protecting fishing grounds from pollution. By encouraging foreign investors in Uganda. By enforcing laws against over fishing. By teaching fishermen the dangers of using poor fishing methods. By encouraging people to practice fish farming. The government should provide fishermen with loans to invest in the fishing industry.				
MINERAL AS A RESOURCE	35	✓ Gives the meaning of a mining, mineral, mineral deposit and mineral ore. ✓ Describes types of minerals ✓ Describes types of mining ✓ Describes the methods of mining ✓ Describes the importance of minerals. ✓ Describes oil mining in the Albertine region. ✓ Describes the contribution of mining industry to Uganda's development. ✓ Discusses the problems facing	**<u>MINERALS AS RESOURCES</u>**. **<u>Mining</u>** is the extraction of minerals from the earth's crust. **<u>A mineral</u>** is a valuable substance found on or below the earth's surface. **<u>A mineral deposit</u>** is a place where a particular mineral exists naturally in large amount. **<u>A mineral ore</u>** is a rock that contains the mineral. **<u>Types of minerals.</u>** Metallic minerals e.g. copper, iron ore, lead, zinc. Non-metallic minerals e.g. Limestone, phosphates, diamonds. Fuel minerals e.g. Coal, crude oil, uranium. **<u>Types of mining.</u>** Traditional mining Modern mining. **<u>Methods of mining</u>**	Guided discovery Guided discussion Think pair share Brain storming		Describing minerals Identifying types of minerals Stating types of minerals	Academic clubs Factory visits Project work making Farming Watering crops

		Uganda's mining industry. ✓ Discusses the possible solutions to the problems facing mining industry. ✓ Discusses the dangers caused by mining industry	Open cast mining e.g. salt, soda ash, limestone, gold. Underground mining e.g. copper, cobalt Drilling method e.g. crude oil Quarrying method e.g. stones, sand. **<u>Importance of some particular minerals</u>.** **Other minrals mined in Uganda.**				Study trips

Mineral	Importance	Area(s) where it exist(s)
Copper	For making electric wires and copper coins, gun bullets.	Kilembe in Kasese
Diamonds	For making jewellery, drilling machines and equipment used for cutting glasses.	Hoima Kamwenge
Gold	For making ornaments, jewellery, trophies, medals.	Kotido in Karamoja. Mubende
Limestone.	For making cement, for making lime.	Hima in Kasese Osukuru hills in Tororo
Salt	For preserving food and human consumption.	Lake Katwe
Petroleum/ Crude oil	For making fuels e.g. petrol, diesel, oil, paraffin.	Shores of lake Albert (Albertine region)
Phosphates	For making artificial fertilizers.	In Tororo
Graphite	For making pencil leads.	Kitgum in Acholi

			Soda ash Wolfram Iron ore Asbestos Tin Graphite. **Note:** *Crude oil was discovered in the Albertine region around the shores of lake Albert.* *An oil refinery is planned in Kaiso-Tonya in Hoima district. Oil will be transported to the coast by the planned Hoima-Tanga crude oil pipeline.* *An oil refinery is a place where crude oil is processed.* *In East Africa, oil is refined at Mombasa and Dar-es-salaam.* ***Examples of oil wells in the Albertine region.*** *Kingfisher oil well on the shores of lake Albert.* *Ngege, Kigogole and Kasememe oil wells in Buliisa.* *Mputa, Nzizi and Ngasa oil wells in Kaiso-Tonya, Hoima.* *Jobi and Lyec oil wells in Nwoya district.* ***How Ugandans will benefit from oil mining in the Albertine region.*** *Oil mining will create job opportunities to people.* *It will promote the development of roads.* *Ugandans will get oil products at a relatively cheaper cost.* *It will lead to development of towns.*	Gypsum Cobalt etc			

			It will promote the development of roads in the region. **<u>Contributions of the Mining industry to Uganda's development.</u>** It is a source of foreign exchange to the government. It is a source of employment opportunities to people. It has promoted the development of roads and railway lines. It leads to urbanisation. It is a source of raw materials for industries. **<u>Problems facing Uganda's Mining industry</u>.** Shortage of capital. Low levels of technology. Shortage of skilled labour force. Unreliable power supply. Mineral exhaustion in some areas. Competition from other sectors e.g. agriculture, fishing etc Poor transport network in different parts of Uganda. **<u>Possible solutions to the problems facing the Mining industry in Uganda.</u>** By constructing better roads to the mining sites. By training more miners. The government should invest more money in the mining industry. By borrowing money from international institutions to invest in the mining industry. By giving tax benefits to foreign investors who invest in the mining industry.				

			<u>Dangers caused by mining</u>. It leads to pollution of the environment. It leads to land degradation. It leads to displacement of people and It promotes soil erosion. animals. It promotes devegetation.				
ANIMAL RESOURCES	36	✓ Describes the types of animals ✓ Describes domestic animals. ✓ Discusses the importance of keeping domestic animals. ✓ Describes wild animals ✓ Discusses the importance of wild animals. ✓ Describes the dangers of animals ✓ Discusses how people mistreat animals. ✓ Discusses ways of caring for animals.	**<u>ANIMAL RESOURCES</u>**. **<u>Types of animals</u>**. Wild animals. Domestic animals. **(i) <u>Domestic animals</u>** are animals kept at home. **Examples are**;- rabbits, sheep, cattle, dogs, goats, camels, donkeys etc. **<u>Importance of keeping domestic animals.</u>** They provide milk eg.cattle, camels, goats Some provide meat eg. Goats, camels, cattle. Some domestic animals provide security/ guard our homes eg.dogs. Some are used for ploughing our gardens eg.oxen Some are used for carrying goods eg.donkeys, camels,horses. Some provide hides and skins. Some domestic animals are used as pets at home. They are a source of income when sold. Some are used for paying brideprice eg cattle, goats.	Guided discovery Guided discussion Think pair share Brain storming		Describing animal resources Identifying importance of keeping domestic animals	Academic clubs Factory visits Project work making Farming Watering crops

			Note: *-Pack animals are animals which are used for transport.* **(ii) <u>Wild animals</u>** are animals that live on their own in their natural habitat. **Examples are;-** snake, lion, zebra, elephant,kob, tiger etc. **<u>Importance of wild animals</u>**. Some wild animals provide meat. Some wild animals provide hooves. Some wild animals provide materials used to make crafts They are used for scientific research Some wild animals are used for cultural purposes. Wild animals attract tourists who bring in income. Some wild animals provide skins and hides. **<u>Dangers of animals.</u>** They destroy farmers' crops. Wild animals like lions, tigers can attack and kill people. Some animals spread diseases to man. Some animals pollute water sources. **<u>How people mistreat animals.</u>** By denying them food. By denying them water.				Study trips

			By overloading pack animals. Through poaching. Through bestiality **<u>Ways of caring for animals</u>**. By feeding animals. By treating sick animals. By giving them shelter. By cleaning their homes.				
CLIMATE AS A RESOURCE	37	✓ Gives the meaning of wind ✓ Discusses the importance of wind. ✓ Discusses the dangers of wind ✓ Describes rainfall as a resource. Describes sunshine as a resource	**<u>WIND AS A RESOURCE.</u>** Wind is air in motion/ moving air. **<u>Importance of wind.</u>** Wind helps in rain formation. Wind helps farmers in winnowing seeds. Winds helps in driving some boats. Wind can be turned into power by wind mills. Wind mills are mainly used in Karamoja to pump water and mill grains. Wind helps in flying kites and balloons. Wind helps our clothes to dry faster. Winds helps in pollution. Wind drives away bad smell. **<u>Dangers of wind.</u>** Strong wind can blow off roofs of our houses. Wind causes soil erosion.	Guided discovery Guided discussion Think pair share Brain storming		Describing wind as a resource Dangers of wind	Academic clubs Project work making Farming Watering crops Study trips

			Strong wind may force boats to capsize on lakes and rivers. Wind pollutes the atmosphere by raising dust. Wind drives away clouds that would bring rainfall. **RAINFALL AS A RESOURCE.** Rain is the main source of water in the environment. **Types of rainfall.** Convectional rainfall (received mostly around forests and large water bodies) Cyclonic rainfall/ Frontal rainfall (received mostly in plateau areas) Relief/Orographic rainfall (received mostly in highland areas) **Importance of rainfall.** Rainfall provides water for domestic use. Rainfall helps plants to grow well. Rainfall reduces dust in the environment. Rainfall increases water volume in water bodies. Rainfall cools the environment. Rainfall softens the soil for easy cultivation. **Dangers of much rainfall.** Much rainfall causes floods. Heavy rainfall causes landslides. It leads to soil erosion. It makes murram roads muddy and slippery. Heavy rainfall leads to destruction of property.				

			Heavy rainfall destroys farmers' crops. **SUNSHINE AS A RESOURCE** The sun is the main natural source of light in the environment. Uganda has receives abundant sunshine throughout the year. This sunshine is tapped and utilized by many people in Uganda in form of solar energy. **Importance of sunshine**. It dries harvested crops. It dries our clothes. It provides light. It helps in generation of solar energy. It is a source of Vitamin D. **Solar energy projects in Uganda.** Kabulasoke solar plant in Gomba. Soroti solar plant at Opuyo-Soroti. Tororo solar plant **Dangers of sunshine.** Strong sunshine leads to drought. It dries up water bodies. It dries pasture for animals. It causes high temperatures in the environment. It causes dust in the envioronment. It dries crops in the garden before they are ready for harvesting				
HUMAN RESOURCE {MAN AS A RESOURCE}	38	✓ Gives the meaning of human resource ✓ Describes types of human labour	**HUMAN RESOURCE (MAN AS A RESOURCE)** Human resource refers to people who are important/ useful to others. **Types of humal labour.**	Guided discovery		Describing labour as a resource	Teaching

		✓ Discusses the importance of people as a resource ✓ Discusses problems faces man as a resource ✓ Discusses ways of improving human resource. ✓ Discusses the problems affecting the utilization of natural resources. ✓ Discusses ways people misuse natural resources. ✓ Discusses ways of protecting natural resources.	**<u>Skilled labour.</u>** This refers to trained workers who provide labour to others. **Forexample;** Teachers Doctors Engineers Bankers Nurses Mechanics Pilots Tailors **<u>Unskilled labour</u>** This refers to untrained workers who provide labour to other people. **Forexample**, Hawkers Potters Market vendors **<u>Importance of people as a resource.</u>** People provide services to other people eg.teachers, doctors etc People help in management of other resources. People provide market for goods and services. People provide labour on farms.	Guided discussion Think pair share Brain storming		Identifying types of labour	Organizing venues Academic clubs Project work making Farming Watering crops Study trips

			People pay taxes to the government. **Problems facing man as a resource.** Poor payment/ low salaries. Illiteracy Poor transport network. Bad weather Poverty Diseases Accidents Political instability. **Ways of improving human resource.** By providing training to workers. By paying attractive salaries to workers. By building better health cenres in the country. By encouraging people to go to school to acquire skills. By building better transport routes in the country. **Problems affecting utilisation of natural resources in Uganda.** Limited capital. Low levels of technology. Shortage of skilled labourforce. Insecurity in some parts of the country. Poor transport and communication. Some natural resources exist in scattered places. **Ways people misuse natural resources.** Through over bush burning.				

			Through over fishing. Through over grazing. Through over cultivation. Through swamp drainage. Through dumping wastes in water bodies. Through uncontrolled harvesting of swamp resources. **<u>Ways of protecting natural resources.</u>** By encouraging people to plant more trees. By teaching farmers better farming methods. By teaching people the dangers of misusing the environment. By enforcing laws againt wastage of resources. By encouraging people to use other sources of energy				